Implementation of a Linux-based Domain Controller with Active Directory and Zabbix Monitoring





Assignment Overview

This assignment explores the real-world application of network management principles through the implementation of a Linux-based Domain Controller with integrated monitoring, aligning with the ISO Network Management Framework.

In this project, a server was configured using Samba to emulate Active Directory (AD) services, enabling centralized user authentication and domain management within a controlled network environment. The setup includes:

- Deployment of a Samba-based AD-compatible domain controller.
- Integration of Kerberos for secure authentication and DNS for name resolution.
- Configuration of at least one client machine (Linux or Windows) to join the domain and authenticate using AD credentials.
- Installation and full configuration of **Zabbix**, a powerful open-source monitoring solution, to oversee the health and performance of both server and client systems.

The Zabbix server was used to create a comprehensive monitoring dashboard, tracking key metrics such as CPU usage, memory performance, disk activity, network status, and the availability of core authentication services. This hands-on project not only reinforces theoretical knowledge but also demonstrates practical skills in network service configuration, domain management, system monitoring, and troubleshooting in a virtualized environment.

The result is a functional mini-enterprise environment that showcases the core competencies expected of a network and systems engineer, bridging the gap between academic learning and professional implementation.

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1. Introduction

In modern network environments, centralized authentication and system monitoring are essential for maintaining security, efficiency and reliability. This project focuses on implementing a Linux-based Domain Controller using Samba to simulate Active Directory (AD) functionalities, enabling centralized user management and domain-based authentication within a local network. In addition to setting up the domain controller, the assignment involves configuring at least one client machine to join the domain and authenticate via AD user credentials.

To complement this setup, the project also integrates Zabbix, an open-source monitoring solution, to ensure visibility into network performance and system health. By combining Active Directory services with network monitoring on a single Linux server, this implementation provides a practical and efficient approach to managing and monitoring small to medium-scale network environments.

This assignment is divided into 3 major parts,

1. Set up a Linux-based domain controller using Samba to implement Active Directory services.
2. Configure a client machine to join the domain and authenticate using the Active Directory user accounts.
3. Install and configure Zabbix to monitor the server and client machine.

2. Resources

- **Operating System**: Ubuntu 22.04 – for Server

  Windows 10 Pro – Client

  Fedora -Client

- **Services**:

  - Samba – to provide Active Directory Domain Controller functionality.
  - Kerberos – for secure authentication within the domain.

- **Monitoring Tool**:

  - Zabbix Server – installed on the same server as the domain controller to monitor system and network performance.

- **Client Machine**:

  - A separate virtual machine running Fedora and Windows 10, joined to the domain for testing authentication and integration.

3. Setting Up the Domain Controller

Configured a Linux-based Domain Controller using Samba to provide Active Directory services for centralized authentication and domain management.

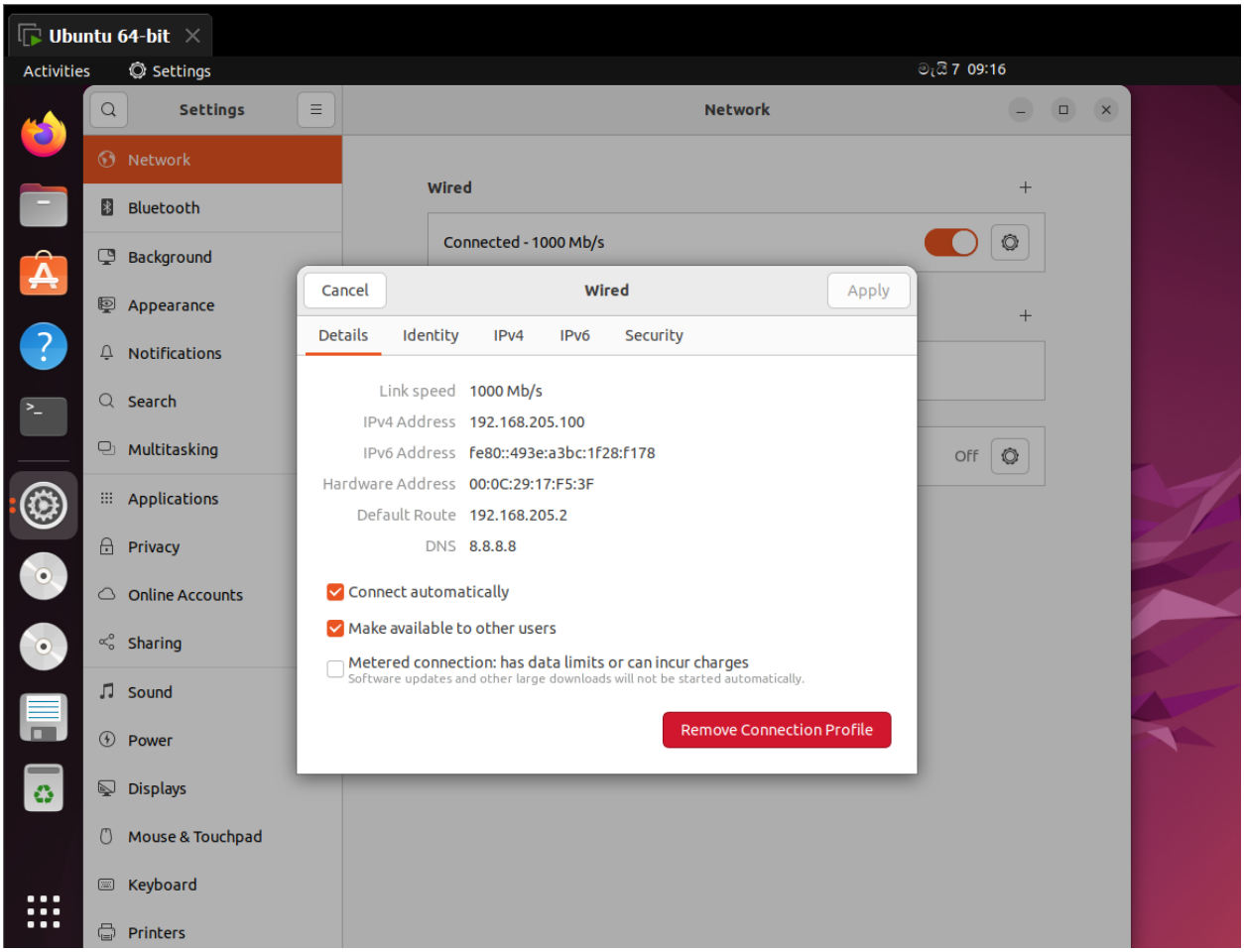


Figure 1.0.1-Edit the Network Configuration on the Ubuntu machine

Active Directory requires a stable and consistent network identity (IP address) for reliable domain services, client communication, and DNS resolution.

- A dynamic IP (from DHCP) can change after a reboot or lease renewal.
- If the IP changes, domain clients won't be able to find or trust the Domain Controller.
- A static IP ensures the DC is always reachable at the same address.

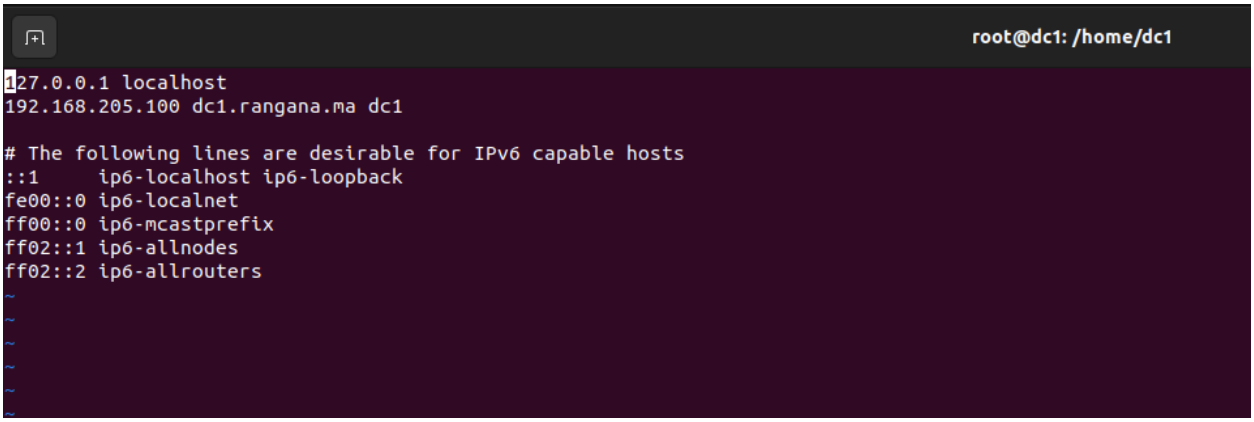


Figure 1.0.2-Add the server to the host file

Add the Samba server IP and domain into the host file(/etc/hosts)

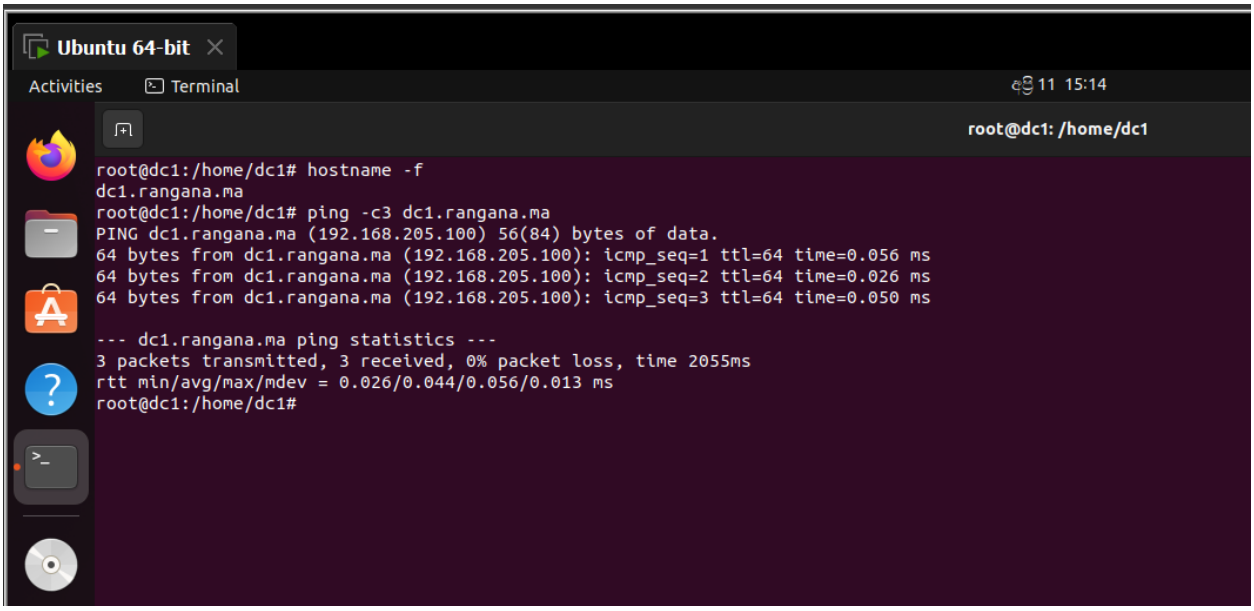


Figure 1.0.3-Verify Hostname

shows the **Fully Qualified Domain Name (FQDN)** of our system and verify that the FQDN resolves to the correct IP address. And also ensures that DNS or /etc/hosts mapping is working properly.

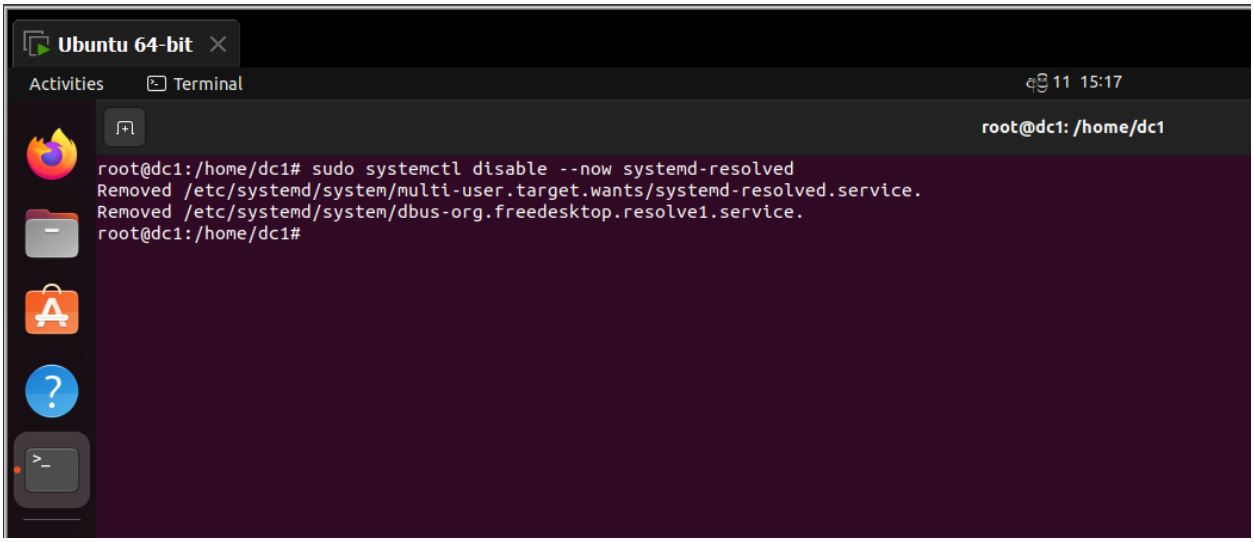


Figure 1.0.4-Disable the DNS Resolver

Because systemd-resolved often overwrites /etc/resolv.conf, which can interfere with custom DNS setups required by Samba AD (like pointing to your own DNS or Domain Controller).

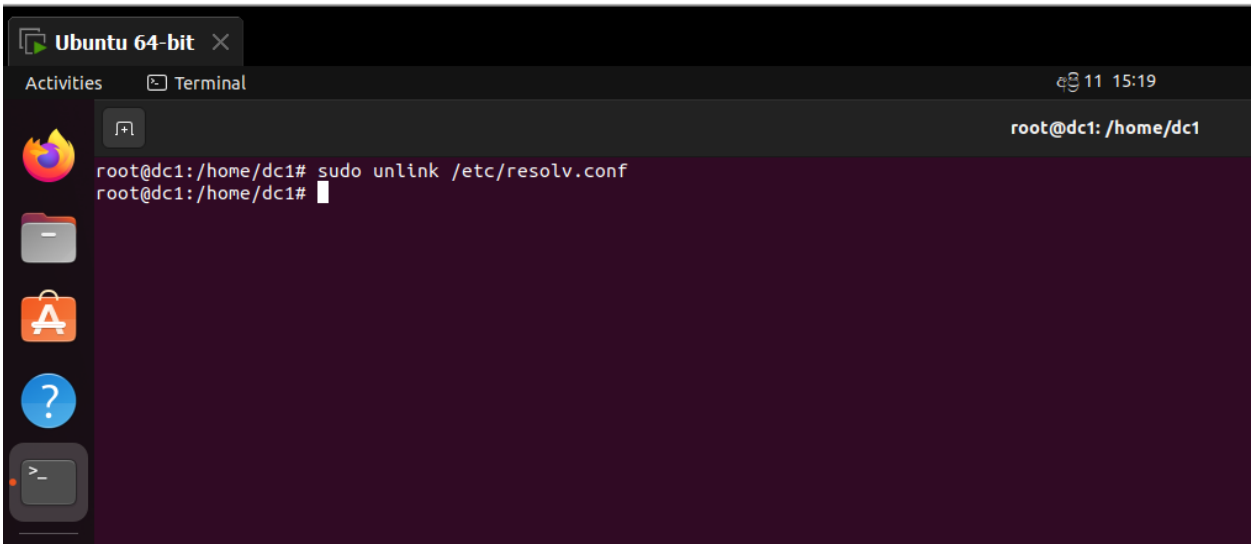


Figure 1.0.5-Unlink the config

- This is necessary so we can manually create your own resolv.conf file with the correct DNS server IP, like the domain controller's IP address.

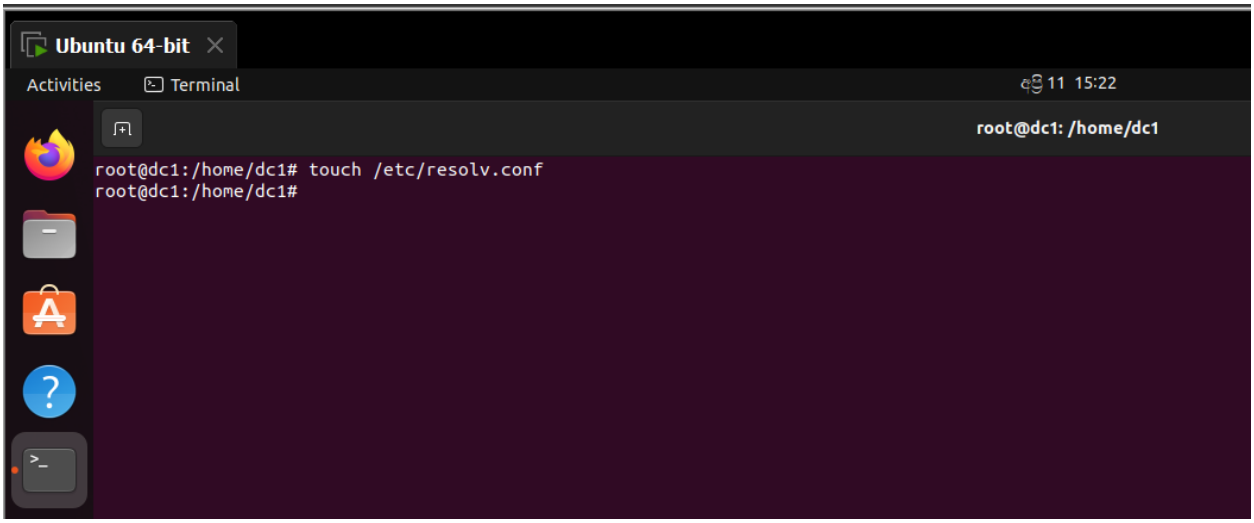


Figure 1.0.6-Create our own Resolv.conf

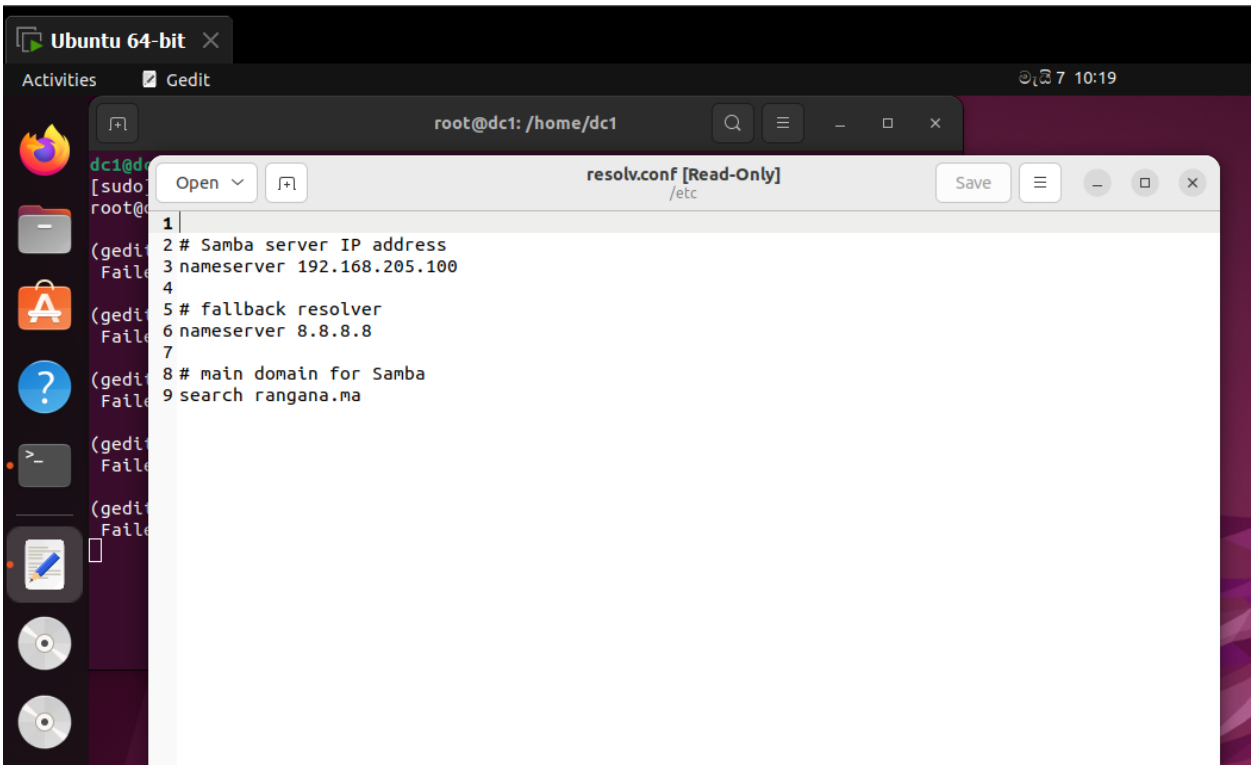


Figure 1.0.7-Edit the newly created resolver config file

This configuration ensures that local domain-related queries go to our Samba DC and internet-related queries fall back to a public DNS, all while using your domain's search path (rangana.ma).

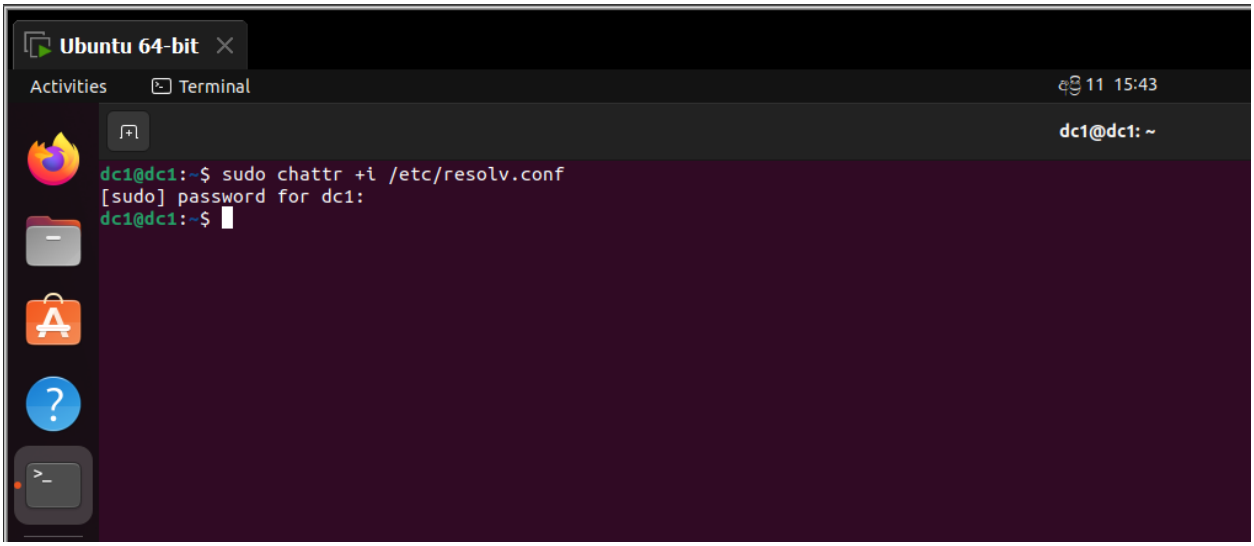


Figure 1.0.8-Make the file immutable

After you manually configure /etc/resolv.conf (with our Samba DC IP as the DNS server), we want to prevent system services like **NetworkManager** or **systemd-resolved** from overwriting it.

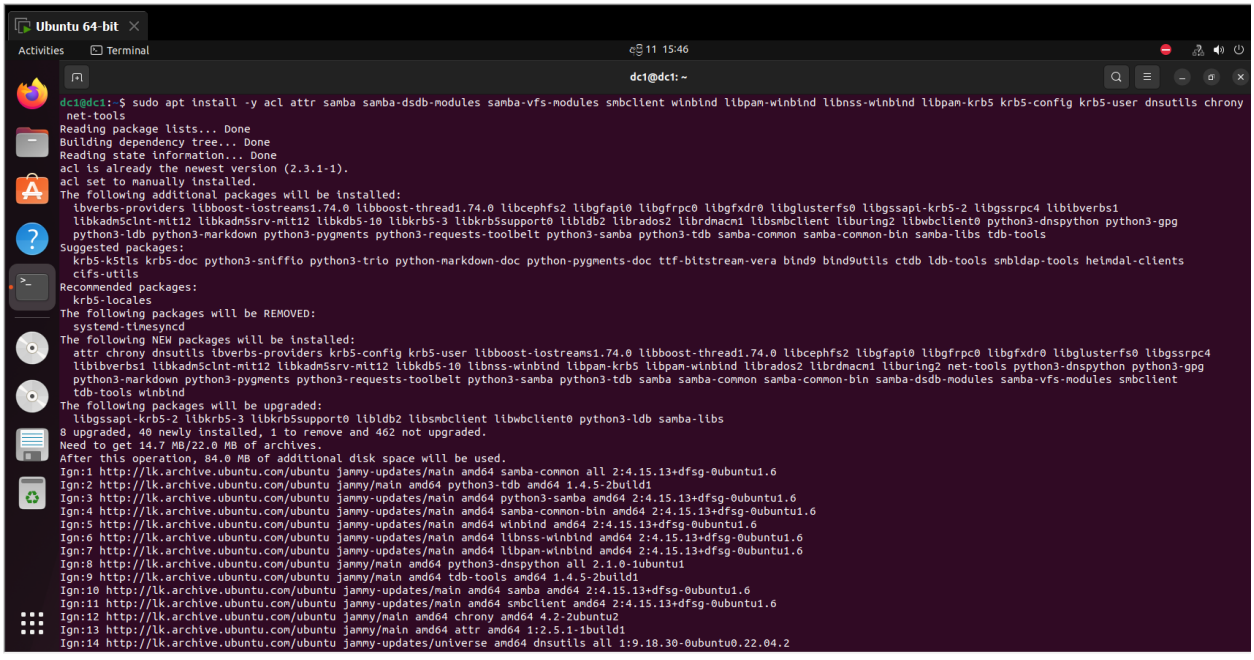


Figure 1.0.9-Install the Samba

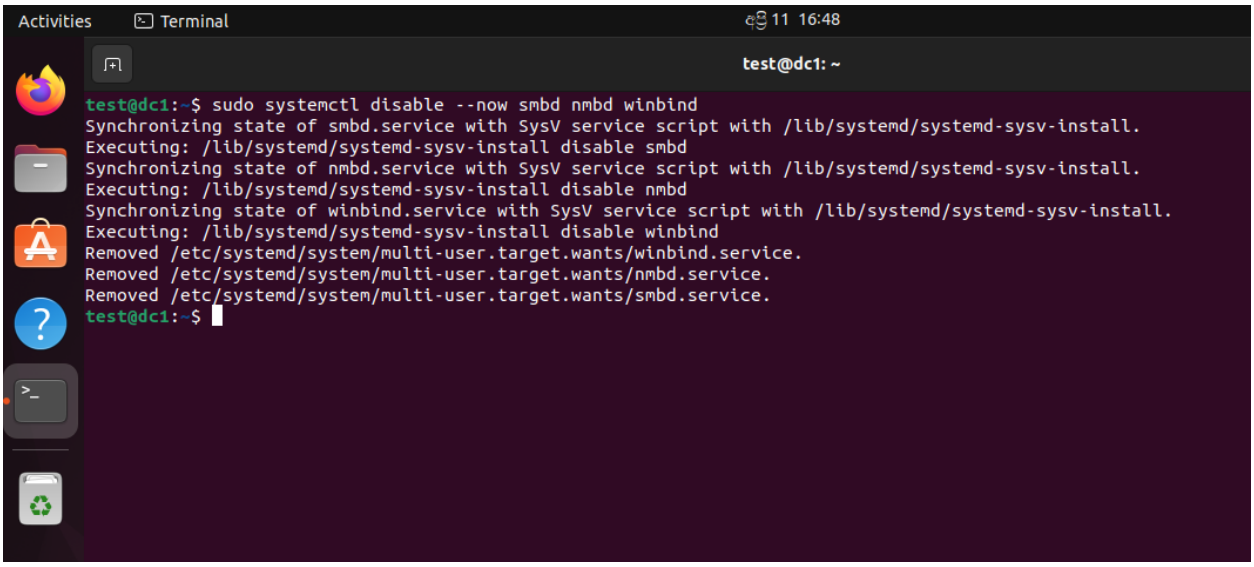


Figure 1.0.10-Disable Samba Services

Stops (--now) and disables the following Samba-related services from starting on boot:

- smbd: Handles SMB/CIFS file sharing.
- nmbd: Manages NetBIOS name service (legacy).
- winbind: Connects Unix/Linux systems to Windows AD for user/group resolution.

You need to stop and disable these services because the Samba AD DC setup uses the Samba service instead, which internally handles all needed functionality. Running smbd, nmbd, or winbind separately can conflict with the Samba service

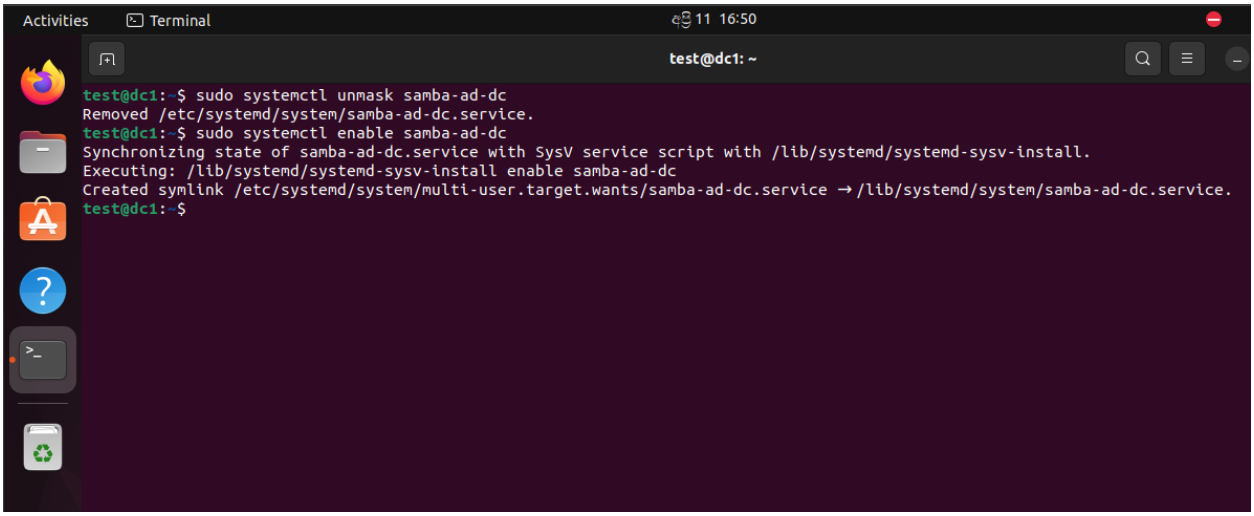


Figure 1.0.11-Active samba-ad-dc

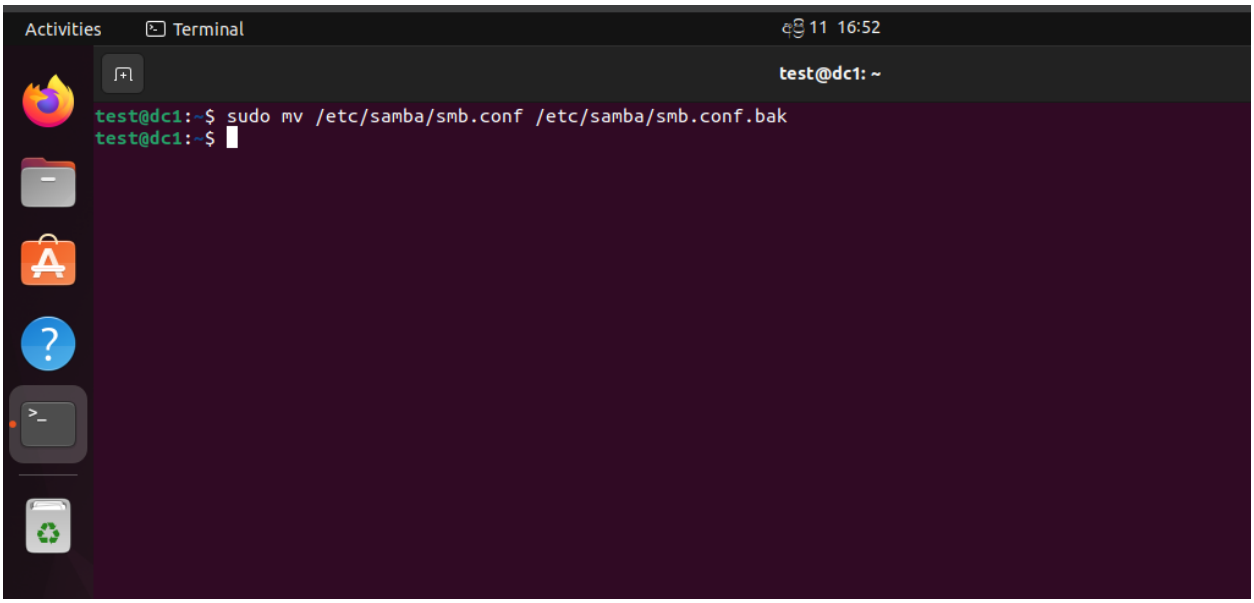


Figure 1.0.12-Back up the original smb.conf file

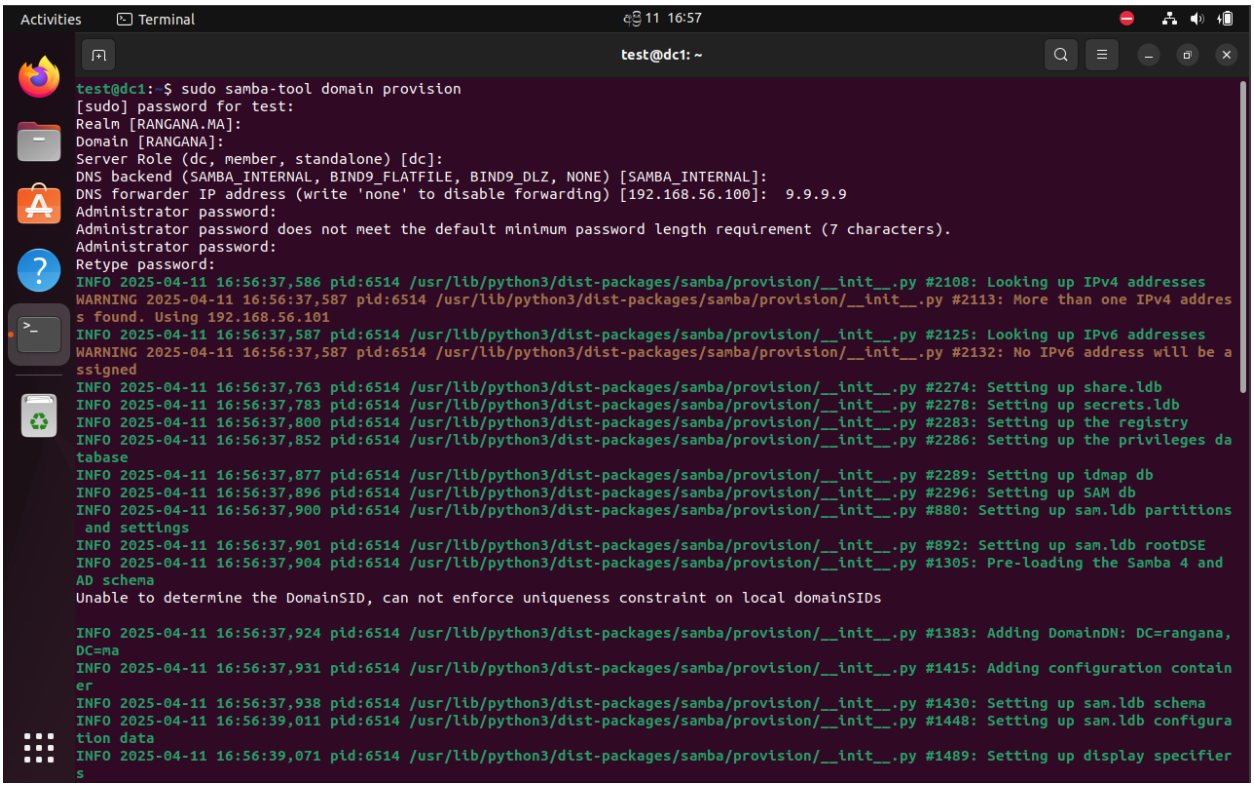


Figure 1.0.13-Provision the Active Directory

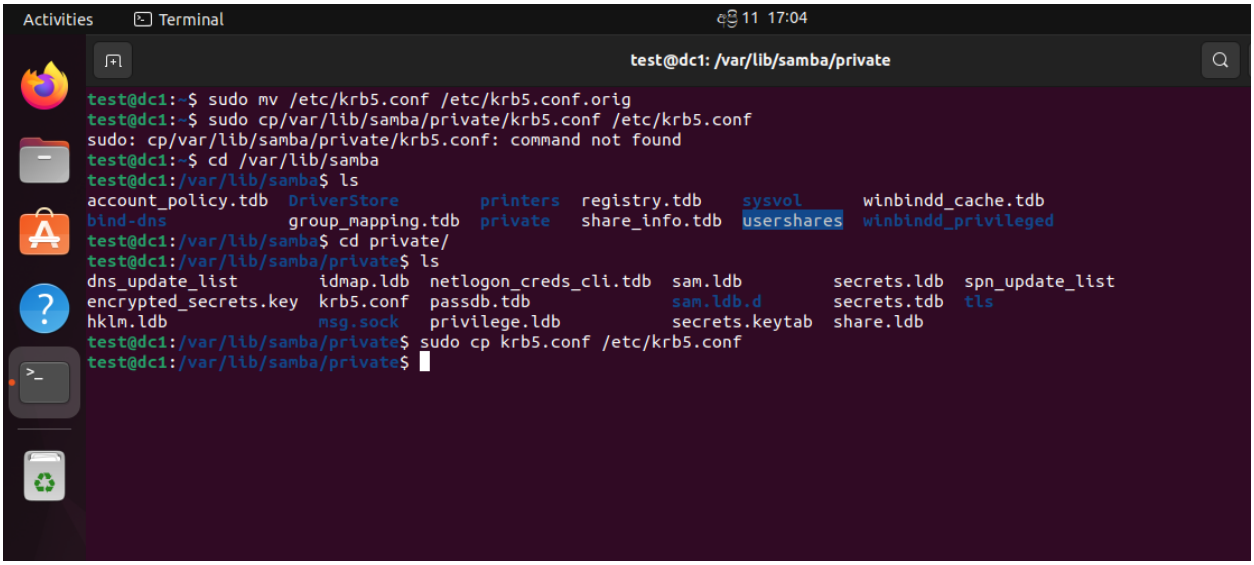


Figure 1.0.14-Backup and replace the Kerberos Config

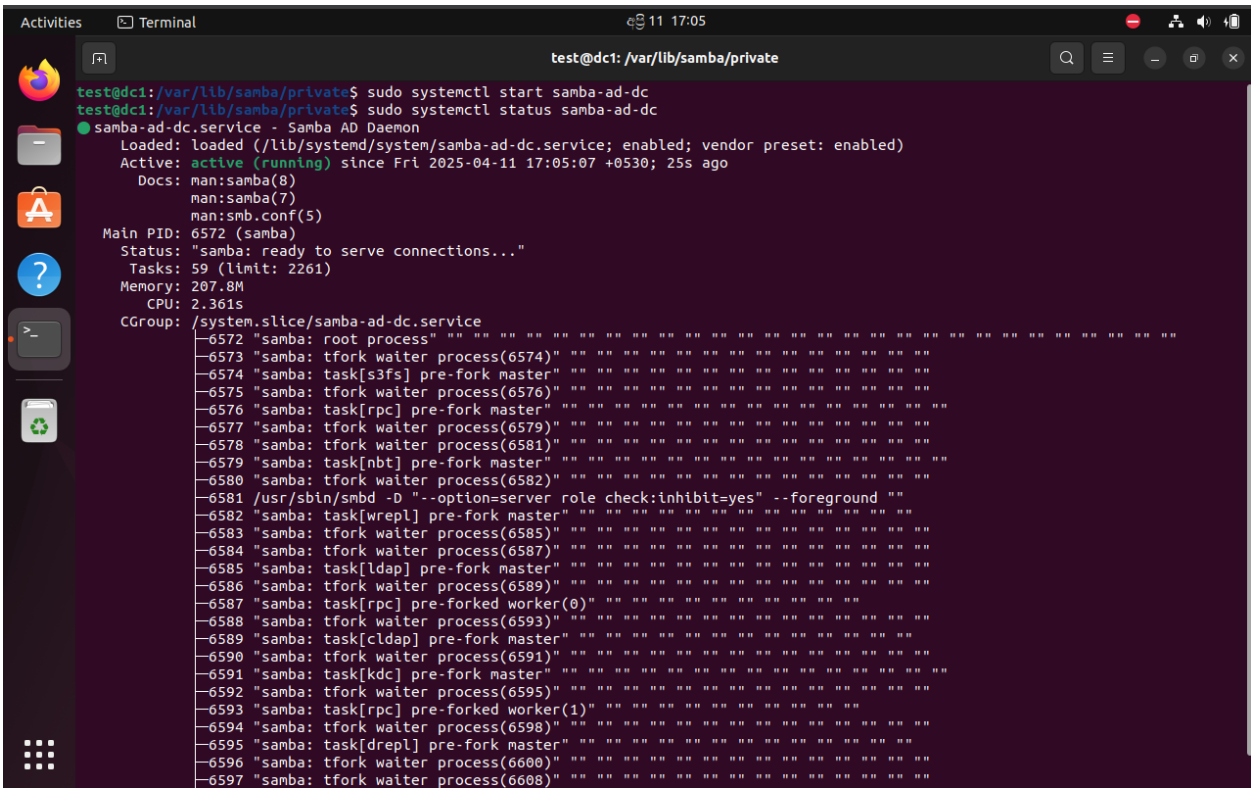


Figure 1.0.15-Start the samba-ad-dc service and check the status

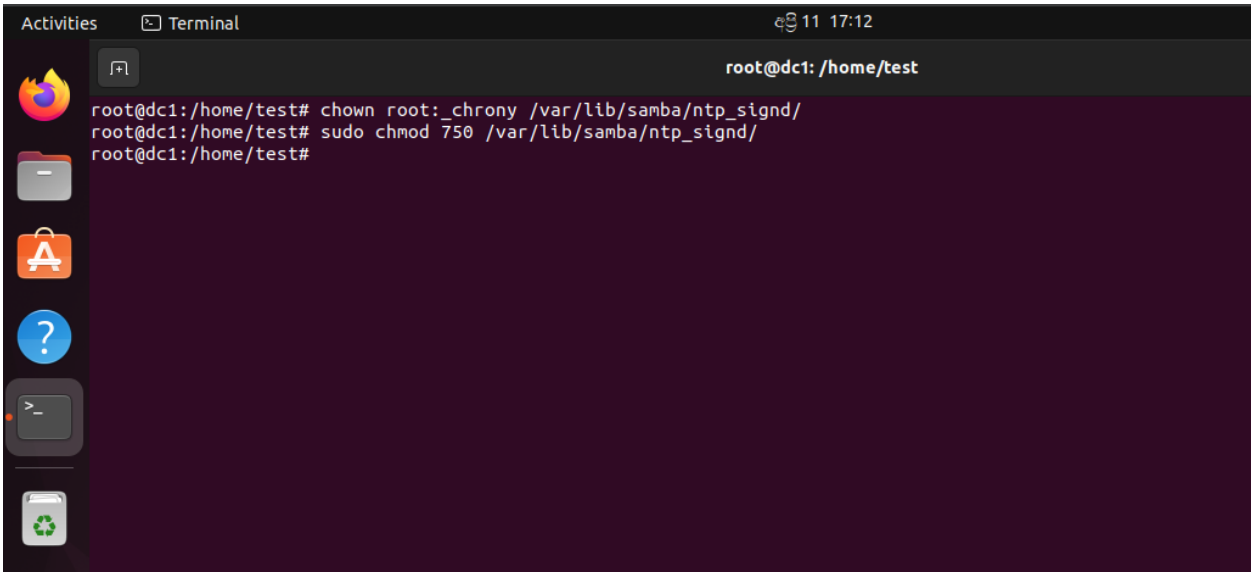


Figure 1.0.16-Set permissions

Changes the ownership of the ntp_signd directory to the _chrony group, allowing the Chrony NTP service to access it.

Sets permissions so that only the owner (root) has full access, the group (_chrony) can read and execute, and others have no access.

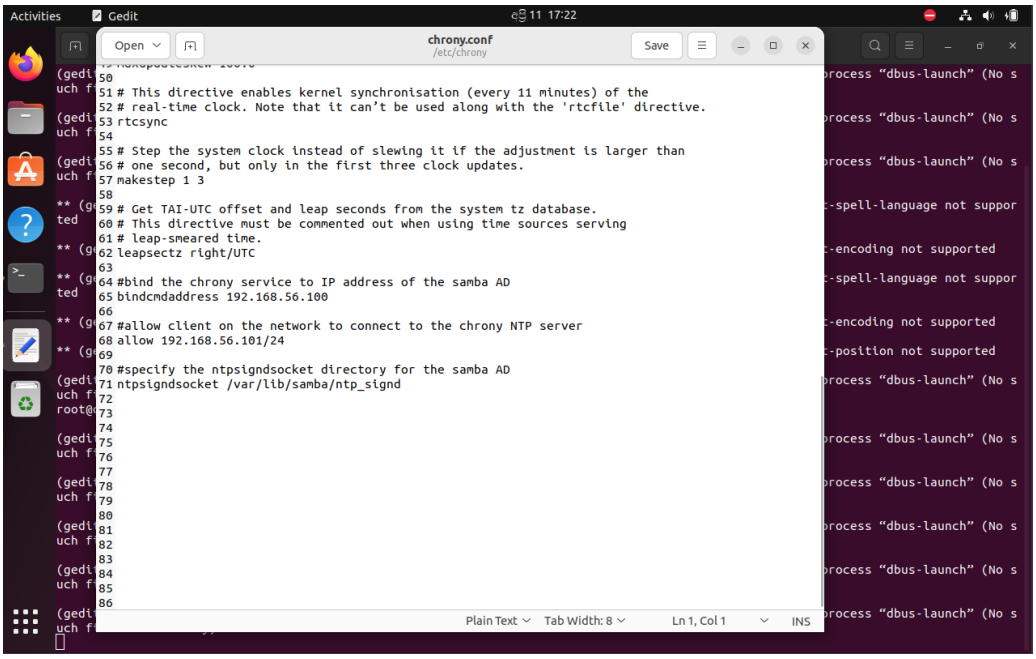


Figure 1.0.17-Update Chrony.conf

Chronyd is the background service of the Chrony time synchronization tool, used to keep a system's clock accurate by syncing with NTP servers. It's faster and more efficient than traditional NTP daemons like ntpd, making it ideal for systems with unstable clocks, such as laptops or virtual machines. chronyd can also act as an NTP server, supports offline correction, and is commonly used in Samba Active Directory environments to provide secure time synchronization for Windows clients.

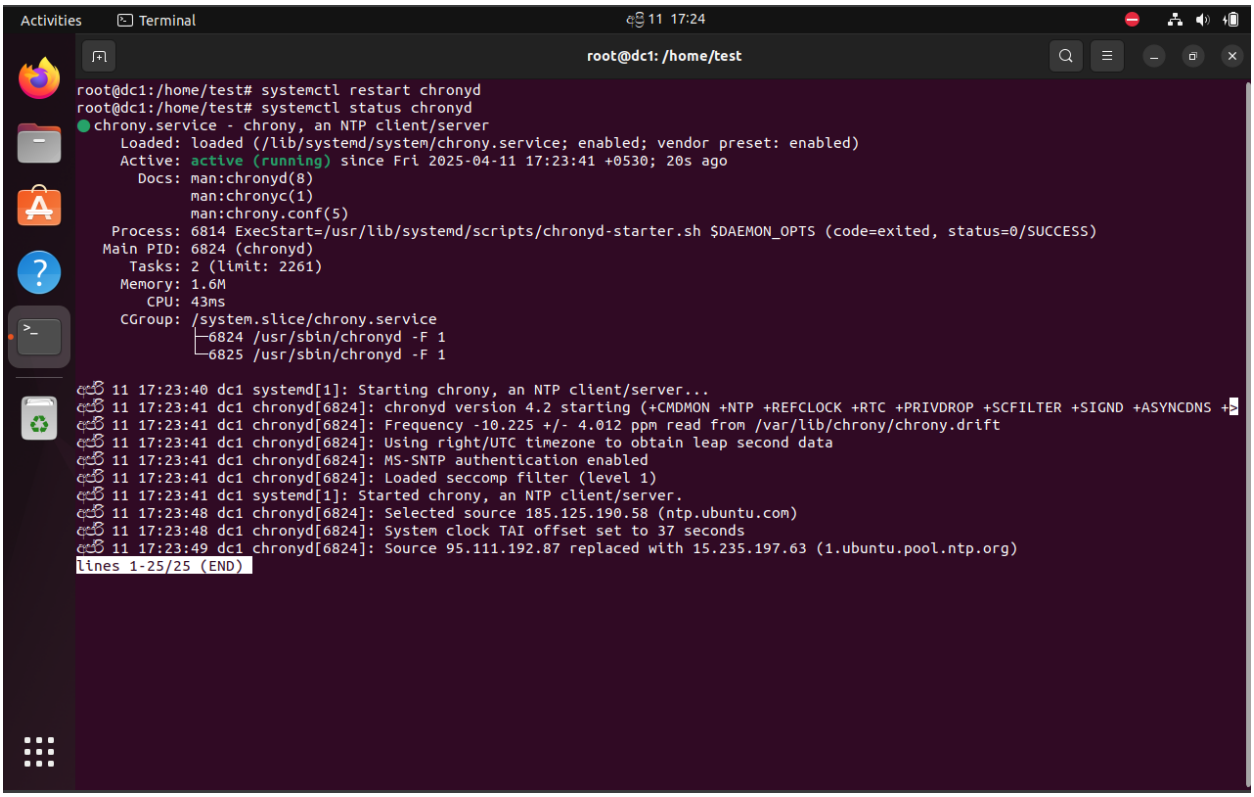


Figure 1.0.18-Restart Chronyd and get its status

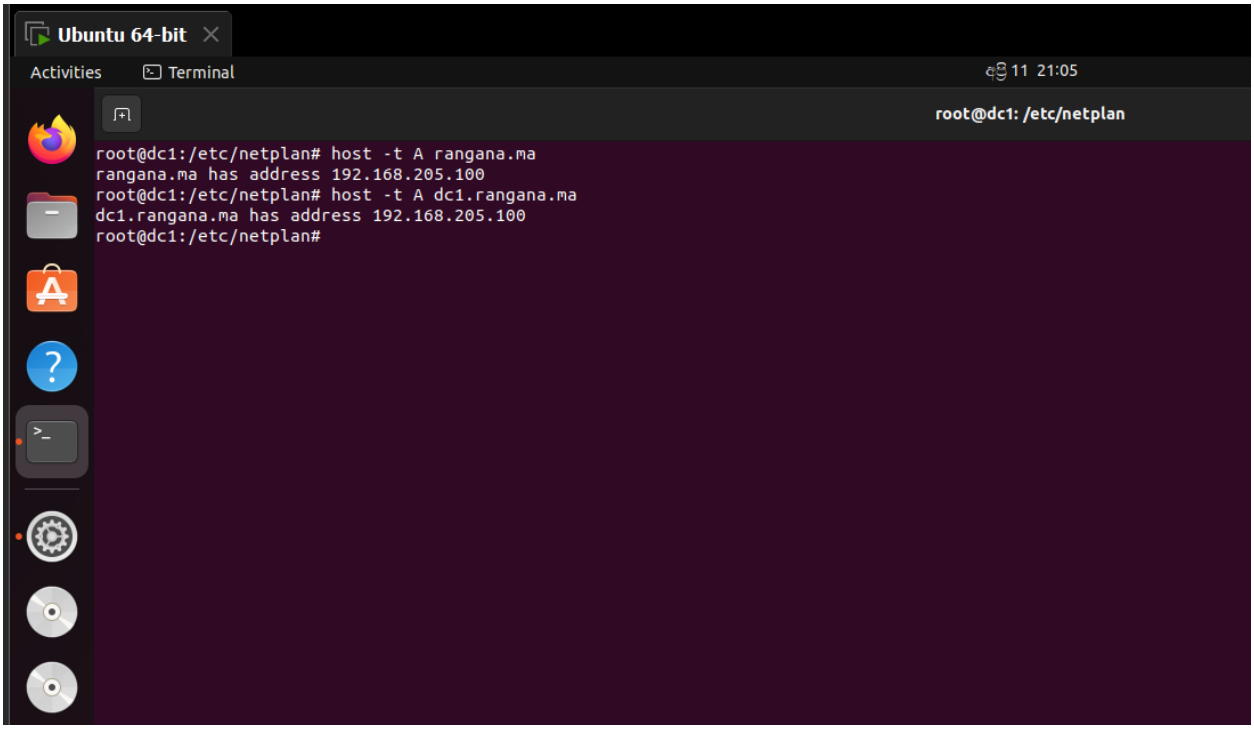


Figure 1.0.19-Verifying Samba Active Directory

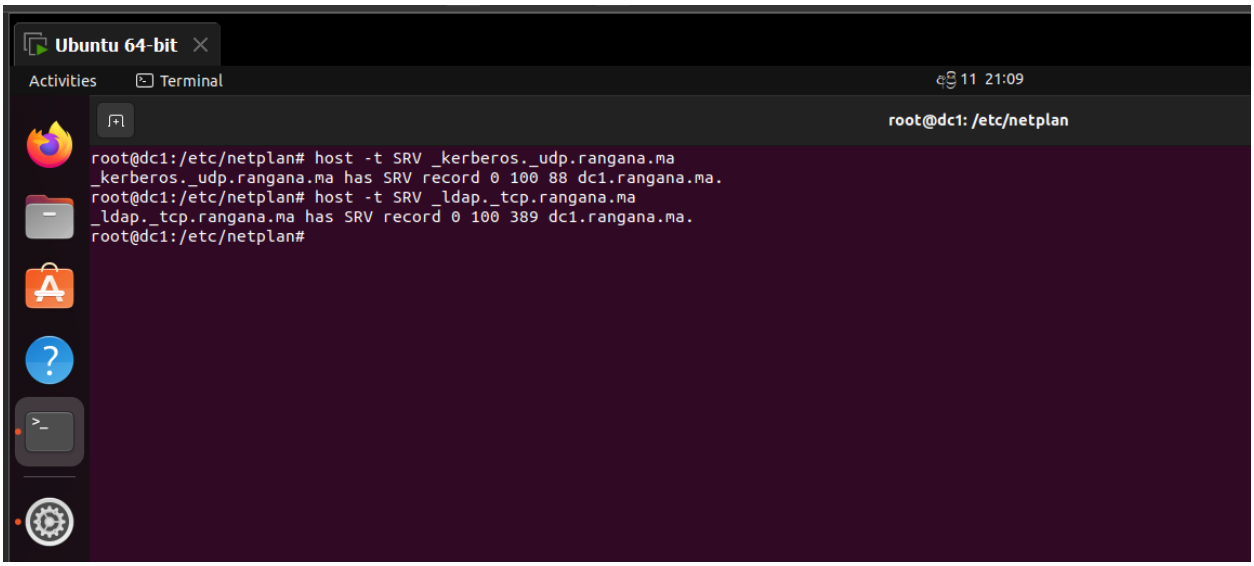


Figure 1.0.20-Verify the Kerberos and ldap services

These commands check if the DNS server correctly advertises essential services for a Samba Active Directory domain. The Kerberos SRV record ensures clients can locate the authentication service, while the LDAP SRV record confirms access to directory services. Proper resolution of these records is required for domain joins and secure client communication with the AD server.

```
root@dc1:~# sudo kinit administrator@RANGANA.MA
Password for administrator@RANGANA.MA:
Warning: Your password will expire in 41 days on 2025-05-23 22:08:09 +0530
root@dc1:~#
```

Figure 1.0.21-Authenticate to Kerberos using an administrator

These commands authenticate the administrator to the Kerberos system (kinit) and then display the cached tickets (klist) to verify successful authentication with the Samba AD domain.

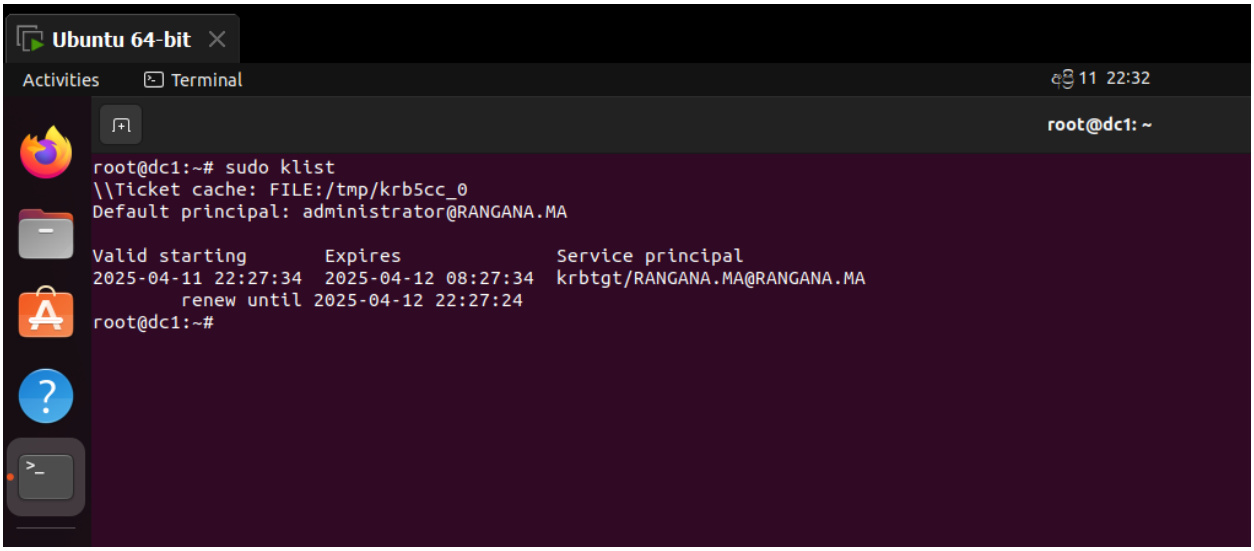


Figure 1.0.22-verify list cached Kerberos tickets

4. Configuring the Client Machine

4.1 Windows Client

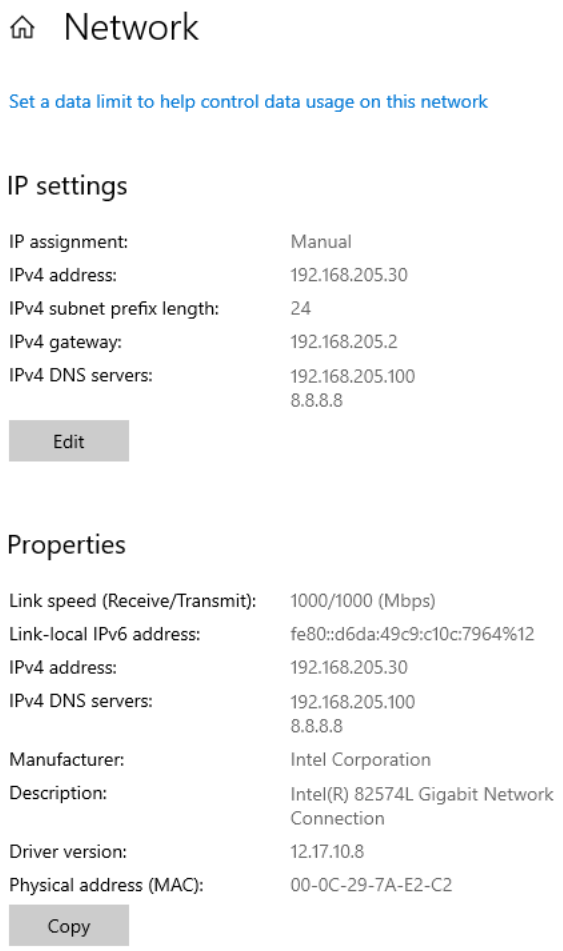


Figure 2.0.1 Edit network Configurations in the Windows machine

We use the same network range IP for the Windows client. The gateway same as the Samba server machine gateway, and DNS is the Samba server IP, and the alternative DNS is 8.8.8.8.

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\malee> Get-DnsClientServerAddress

InterfaceAlias               Interface Address ServerAddresses
                             Index     Family
--------------               --------- ------- --------------
Ethernet0                           12 IPv4    {192.168.205.100, 8.8.8.8}
Ethernet0                           12 IPv6    {}
Loopback Pseudo-Interface 1          1 IPv4    {}
Loopback Pseudo-Interface 1          1 IPv6    {fec0:0:0:ffff::1, fec0:0:0:ffff::2, fec0:0:0:ffff::3}
```

Figure 2.0.2-Verify the DNS resolver

```
PS C:\Users\malee> ping dc1.rangana.ma

Pinging dc1.rangana.ma [192.168.205.100] with 32 bytes of data:
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.205.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\malee> ping rangana.ma
```

```
PS C:\Users\malee> ping rangana.ma

Pinging rangana.ma [192.168.205.100] with 32 bytes of data:
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64
Reply from 192.168.205.100: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.205.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\malee>
```

Figure 2.0.3-Ping AD domain dc1.rangana.ma and rangana.ma

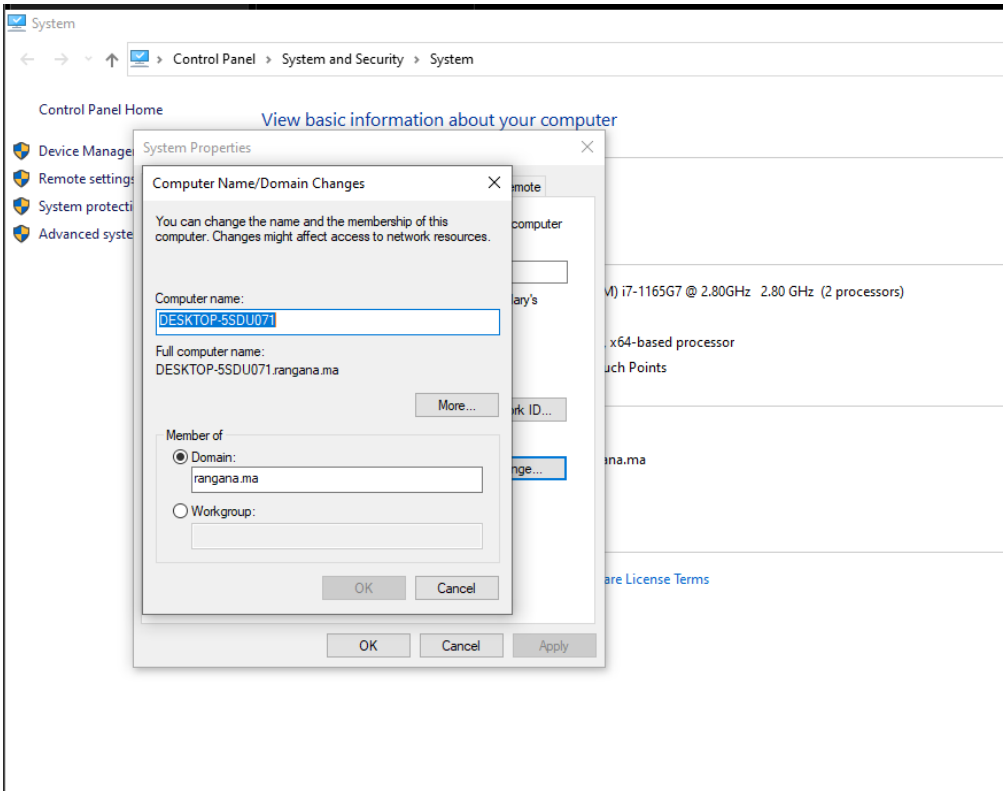


Figure 2.0.4-Add Windows 10 to Active Directory

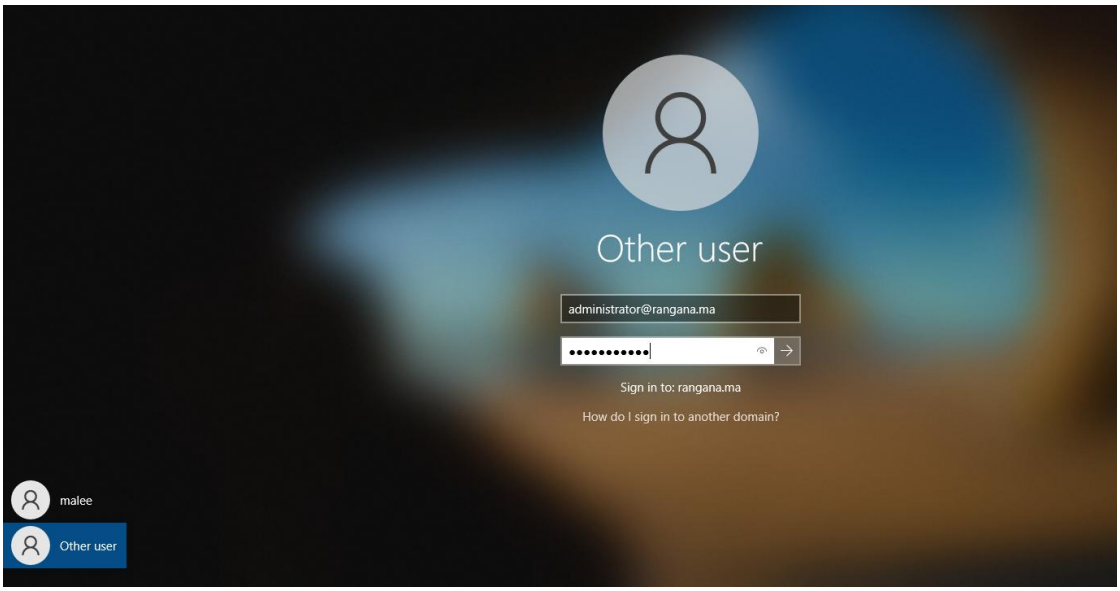


Figure 2.0.5-Logging in with the administrator account

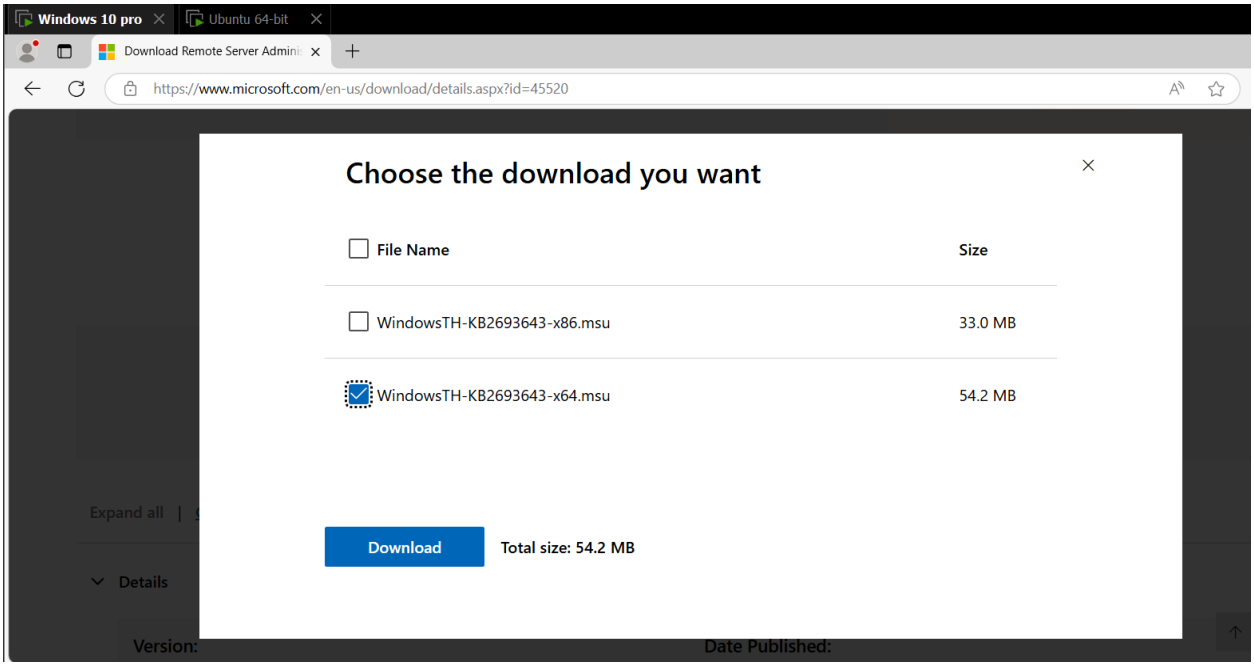


Figure 2.0.6-Download RSAT to administer the domain from Windows

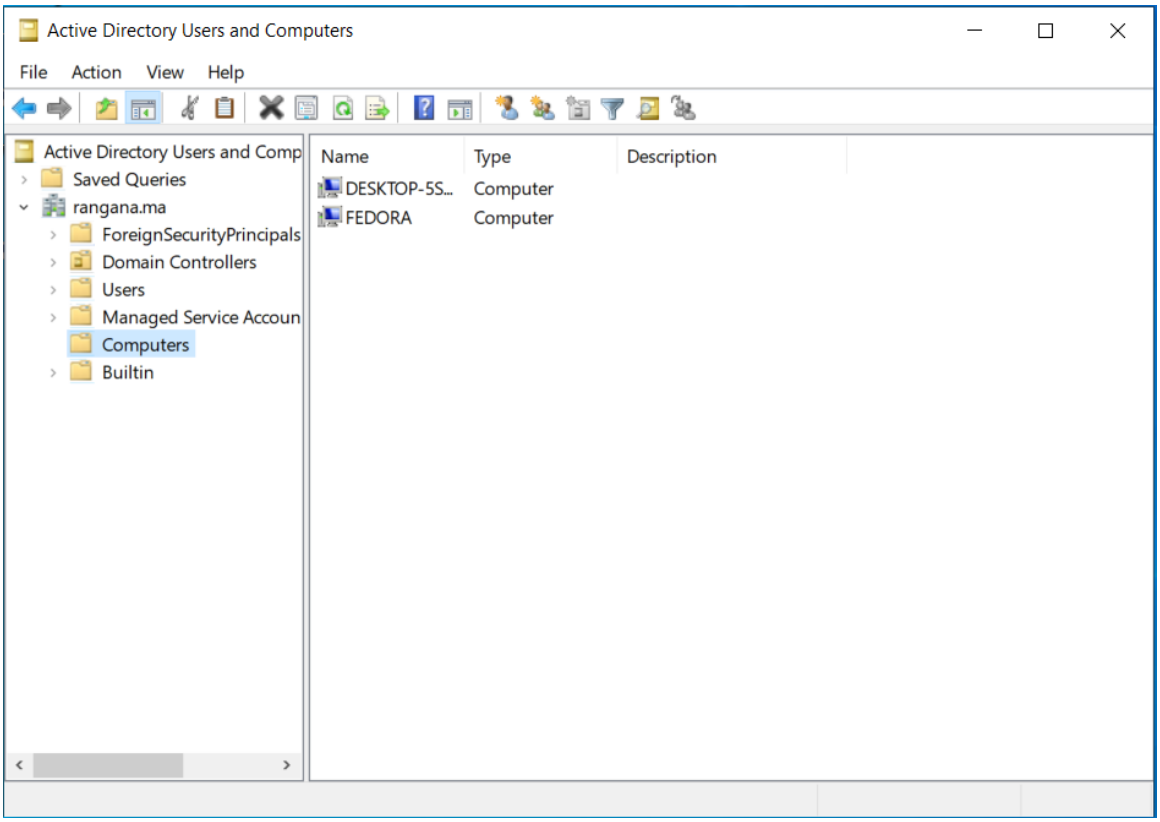


Figure 2.0.7-AD user and computers

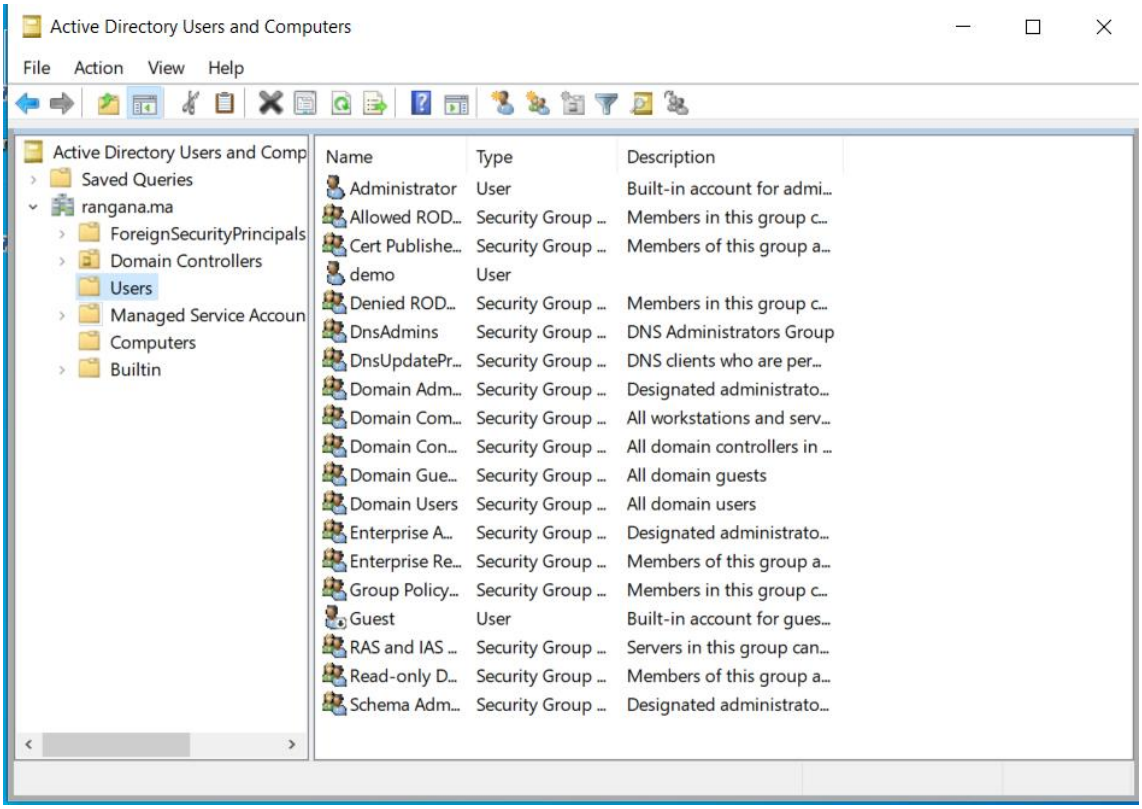


Figure 2.0.8-User's window

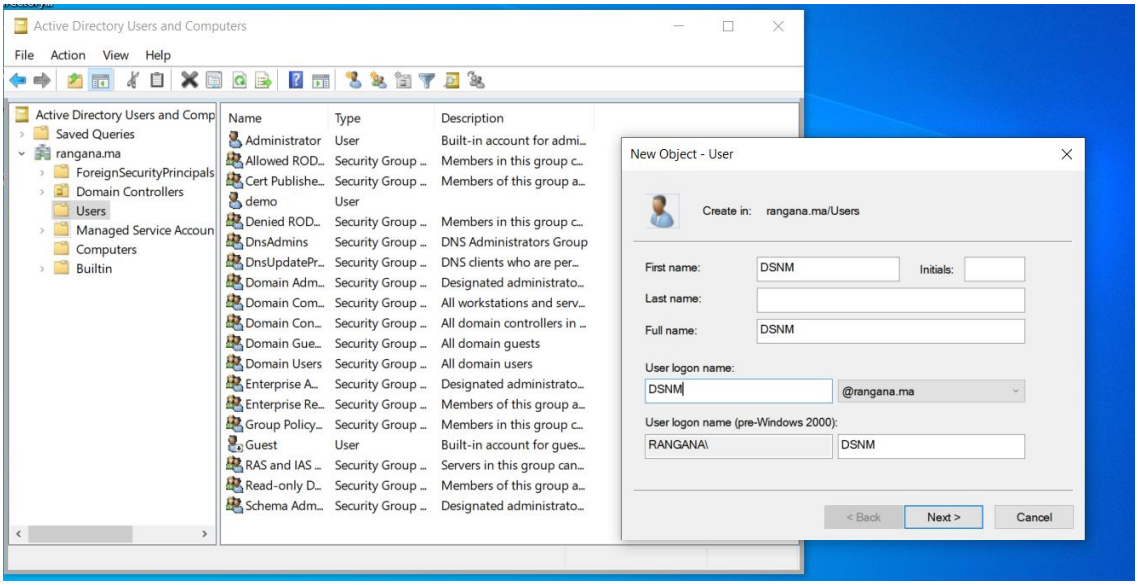


Figure 2.0.9-Create a new user called DSNM

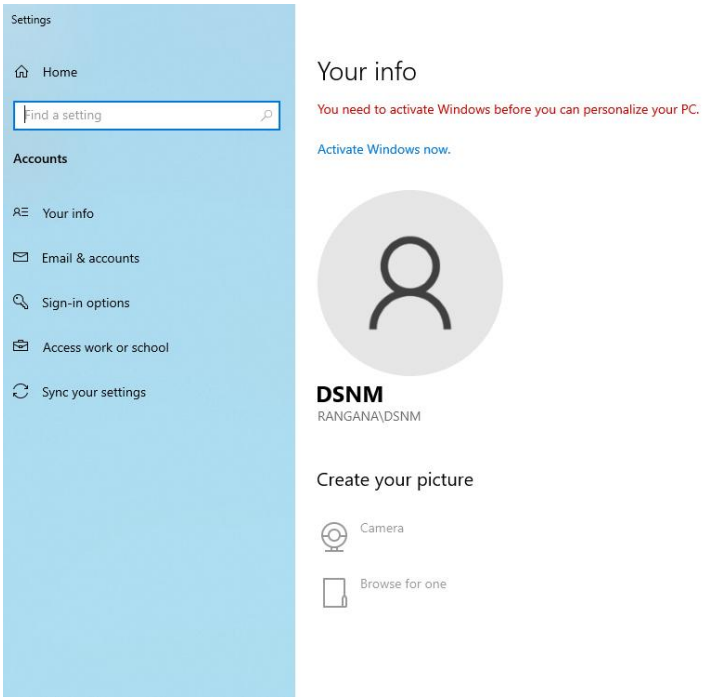


Figure 2.0.10-Logged in DSNM user Account

4.2 Fedora Client

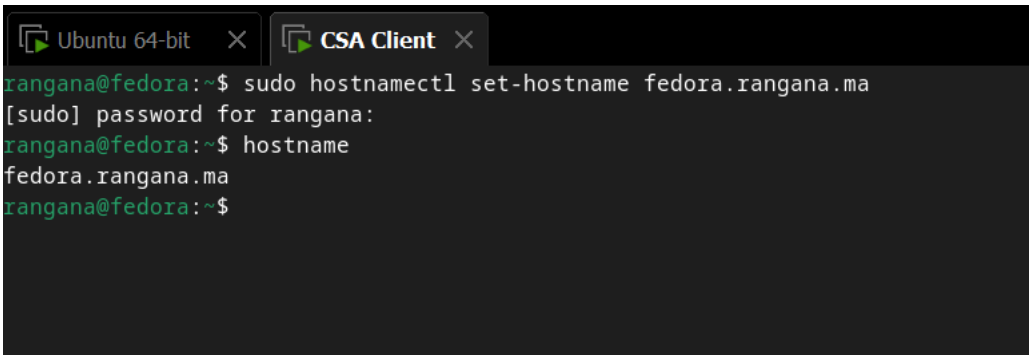


Figure 2.0.11-Change hostname

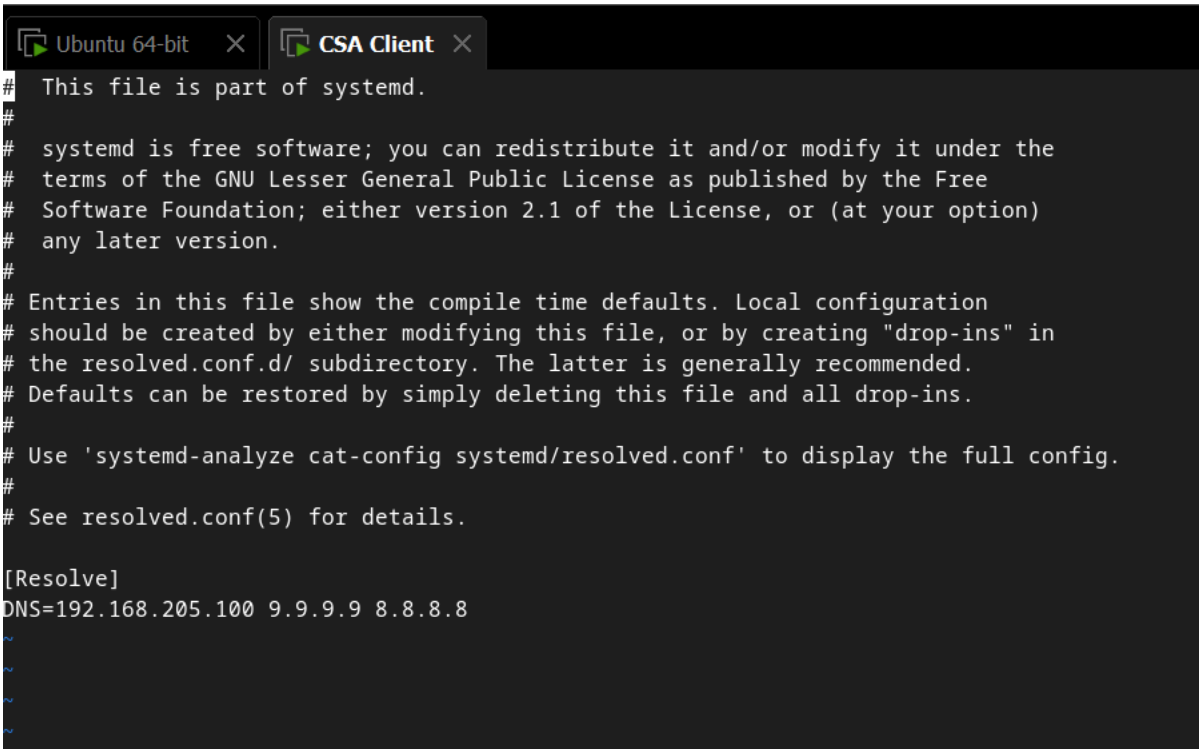


Figure 2.0.12-Modify resolved information

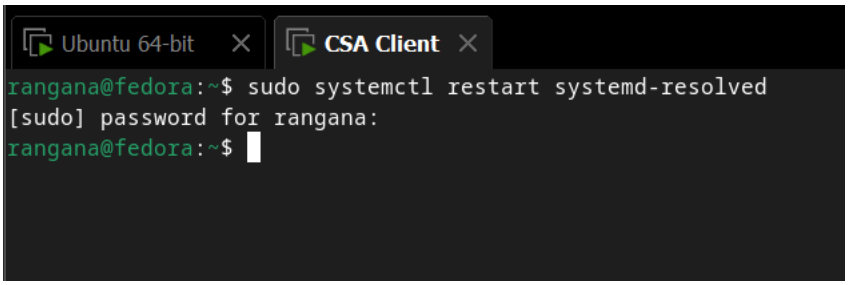


Figure 2.0.13-Restart systemd-resolved

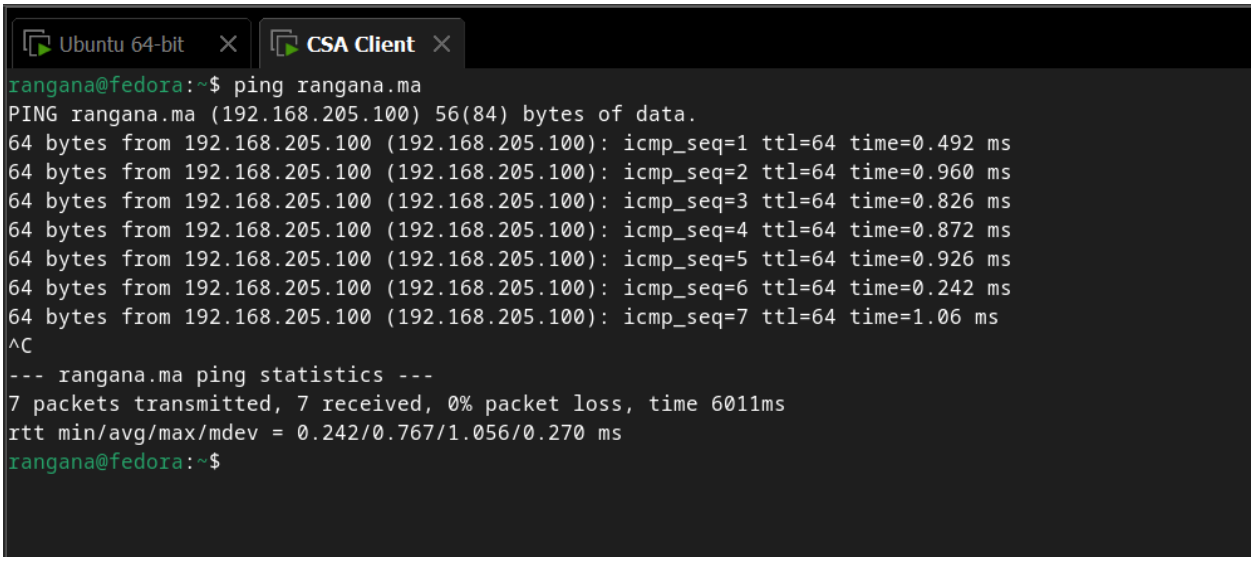


Figure 2.0.14-Ping the Samba domain

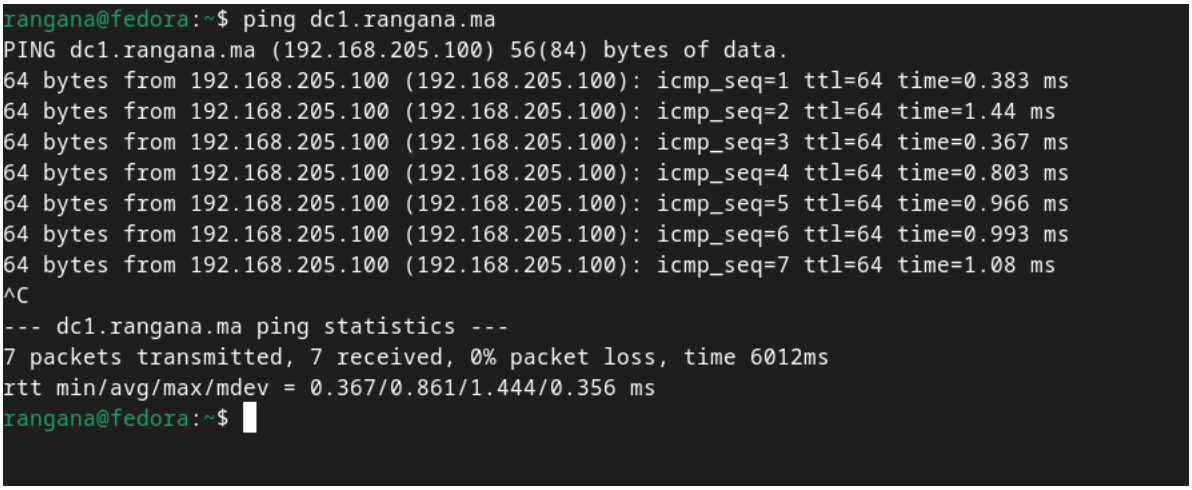


Figure 2.0.15-Ping dc1.rangana.ma

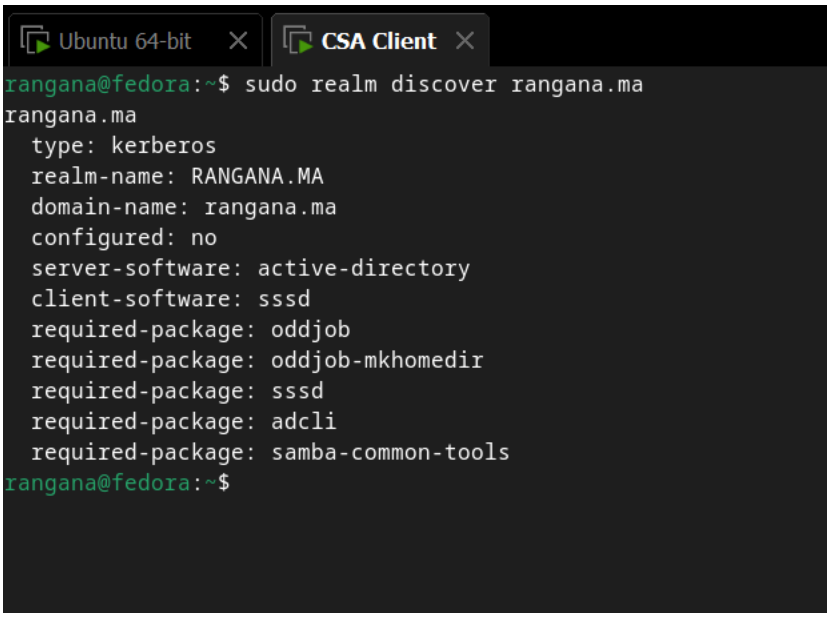


Figure 2.0.16-Discover the Domain

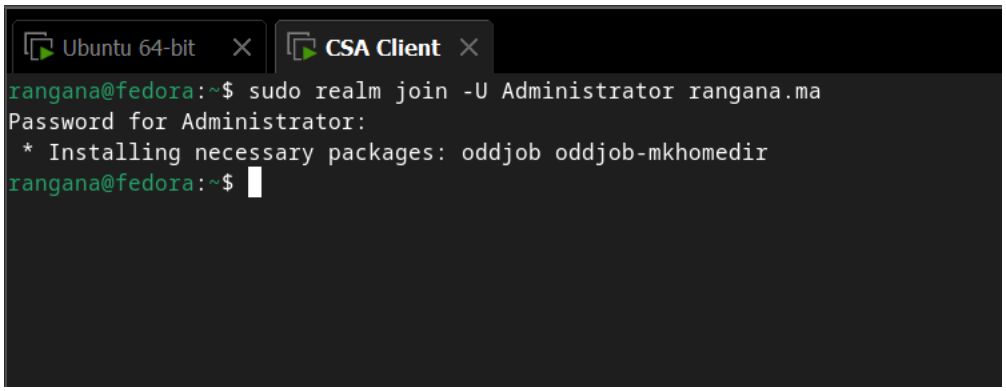


Figure 2.0.17-Join the Domain



Figure 2.0.18-Log in with the created demo user in Fedora

5. Install and configure Zabbix

This setup installs and configures Zabbix 7.0 LTS on Ubuntu 22.04 (Jammy) with support for Server, Frontend, and Agent components. It uses Mysql as the database and Apache as the web server, providing a stable and long-term supported monitoring solution for networks and systems.

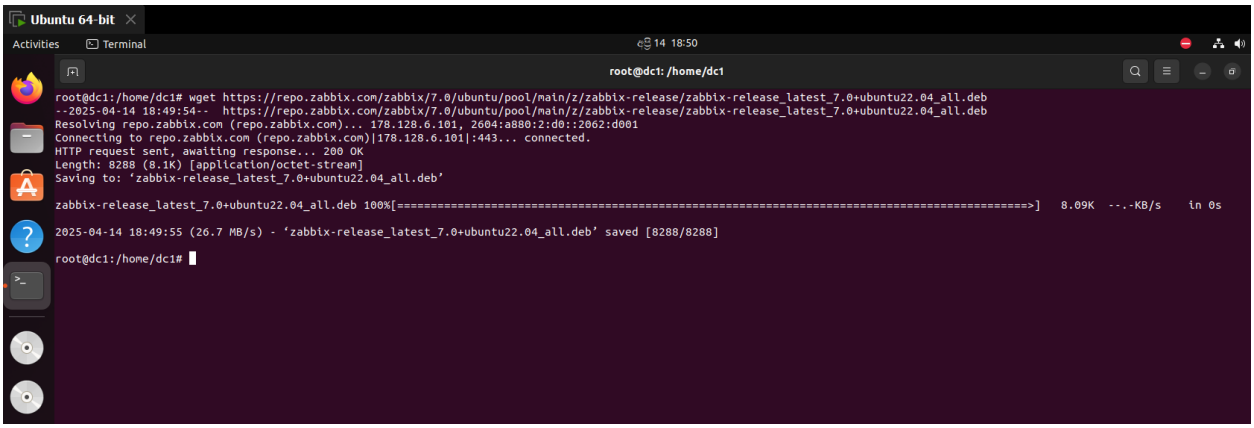


Figure 3.0.1-Downloads the latest Zabbix repository .deb package for Ubuntu 22.04

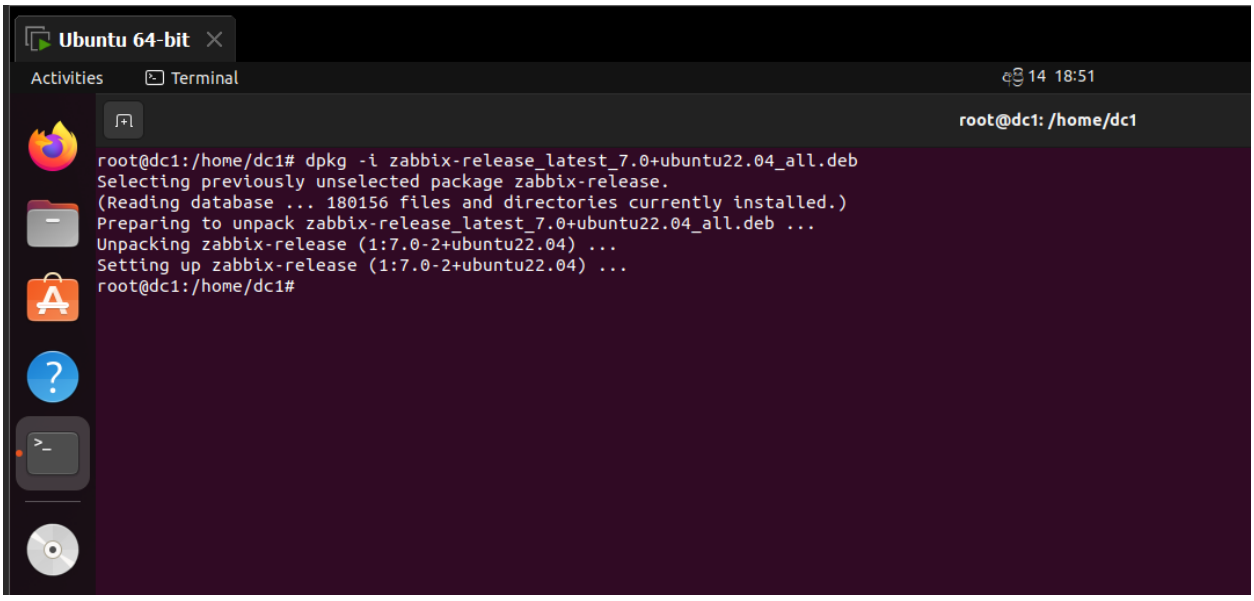


Figure 3.0.2-Installs the downloaded .deb file, which adds the Zabbix APT repository

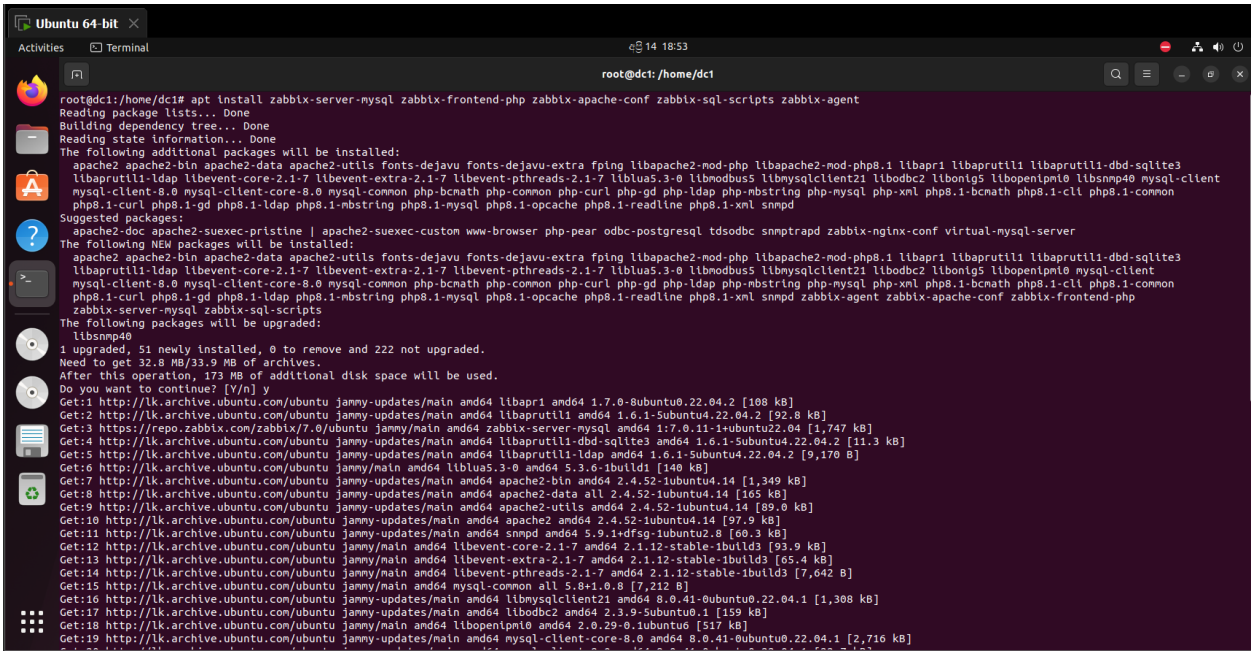


Figure 3.0.3-Install Zabbix server, frontend, agent

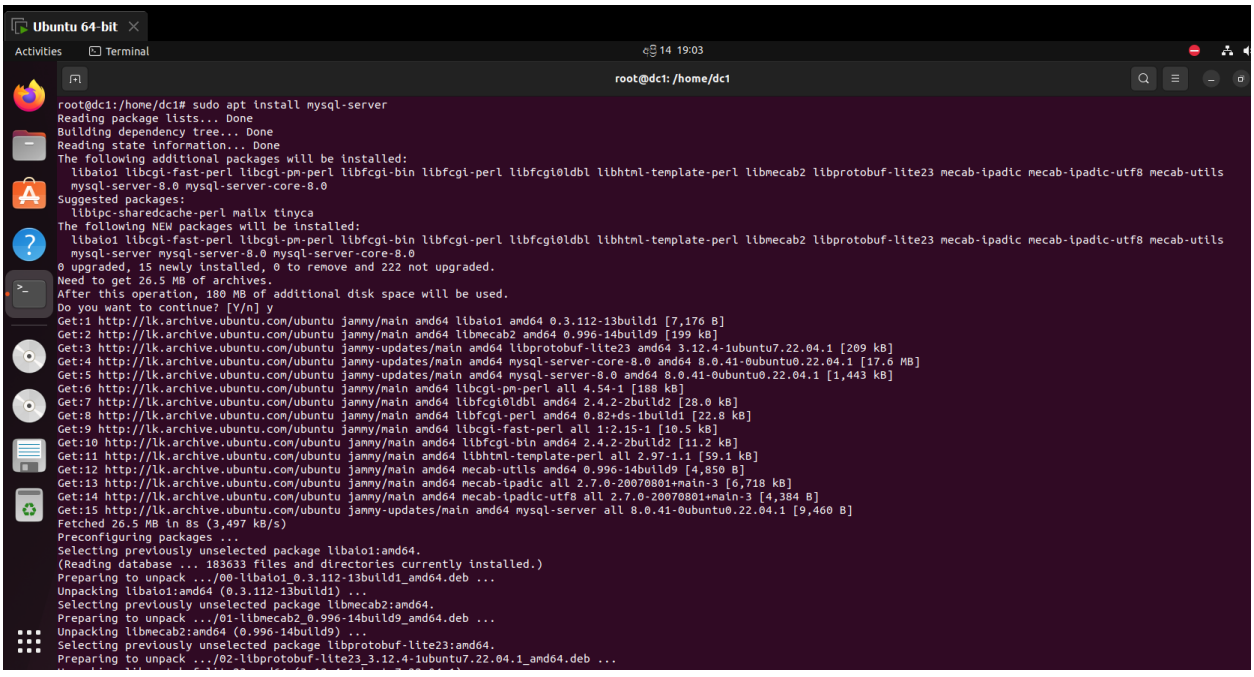


Figure 3.0.4-Install Mysql server

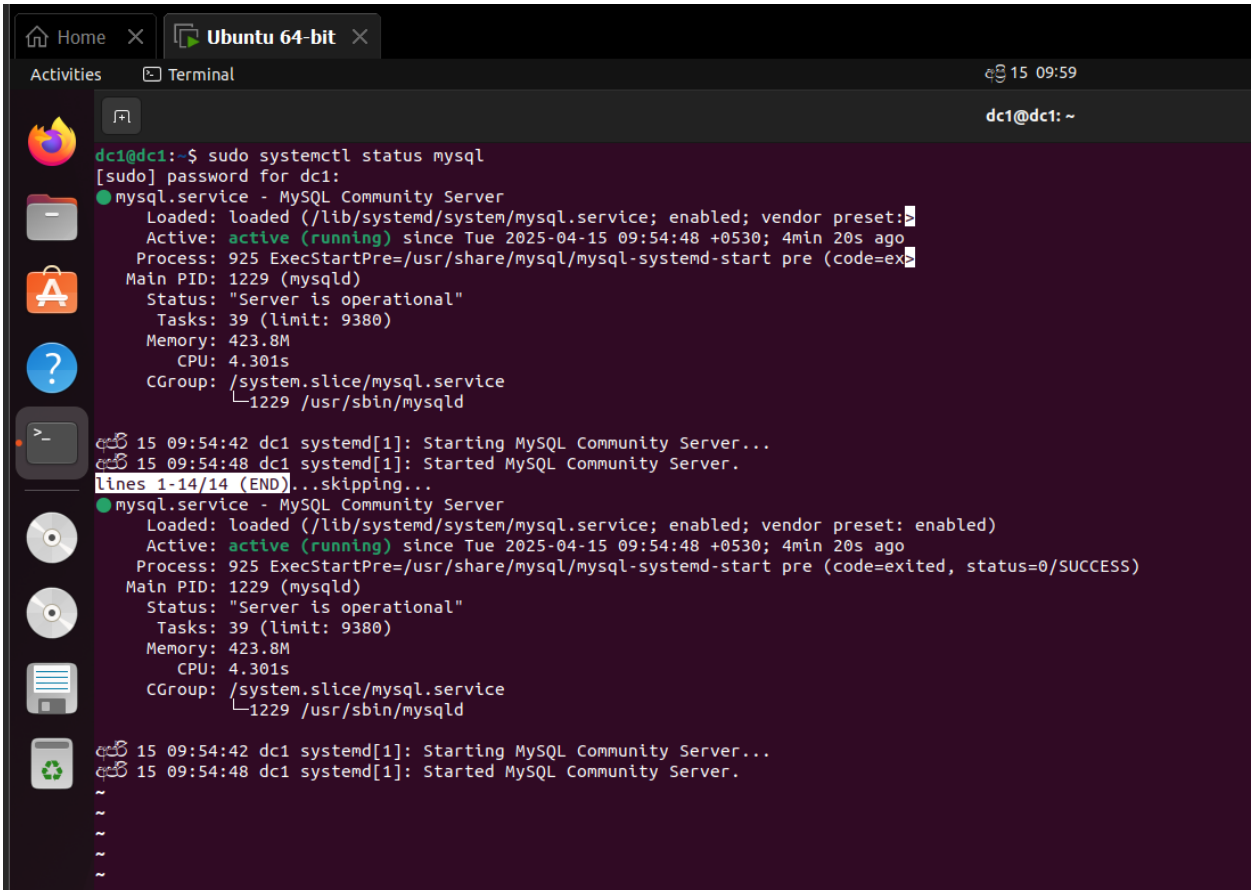


Figure 3.0.5-mysql status

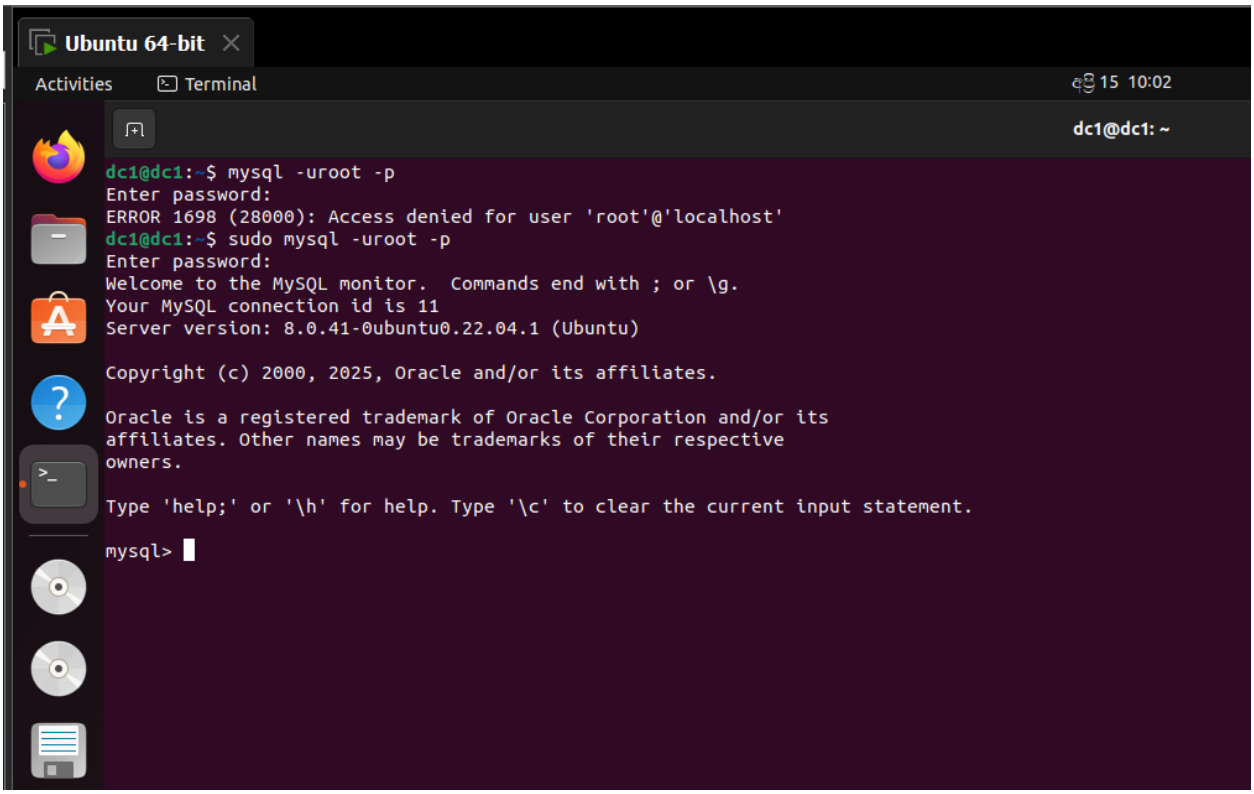


Figure 3.0.6-Loggin mysql server as root user

```
mysql> create database zabbix character set utf8mb4 collate utf8mb4_bin;
Query OK, 1 row affected (0.00 sec)
```

Figure 3.0.7-Creates a new database named zabbix

```
mysql> create user zabbix@localhost identified by '123';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Figure 3.0.8-Creates a new user called zabbix.

```
mysql> grant all privileges on zabbix.* to zabbix@localhost;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Figure 3.0.9-Gives the Zabbix user full access to the Zabbix database only

```
mysql> set global log_bin_trust_function_creators = 1;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> quit
Bye
dc1@dc1:~$
```

Figure 3.0.10-Create stored functions.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| zabbix             |
+--------------------+
5 rows in set (0.01 sec)

mysql>
```

Figure 3.0.11-Show databases

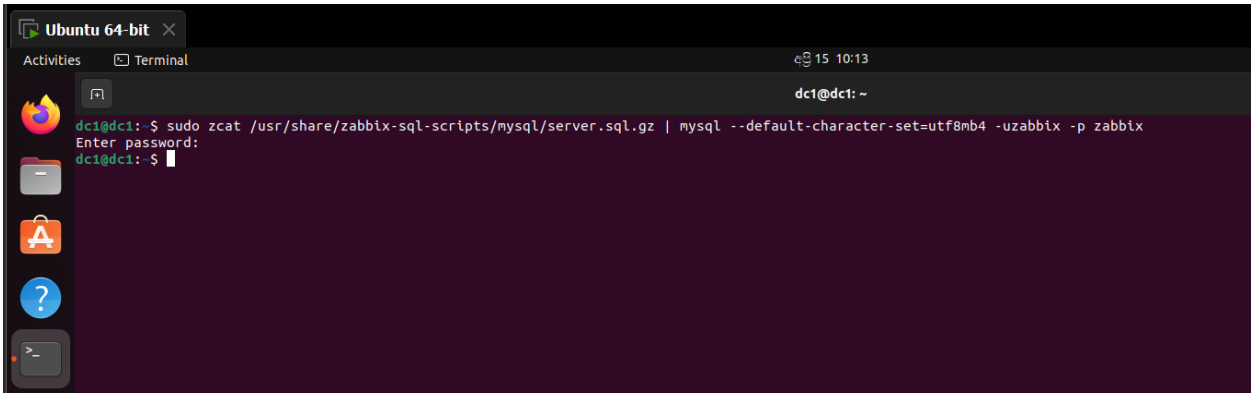


Figure 3.0.12-Import the Zabbix database structure and data into the Mysql database

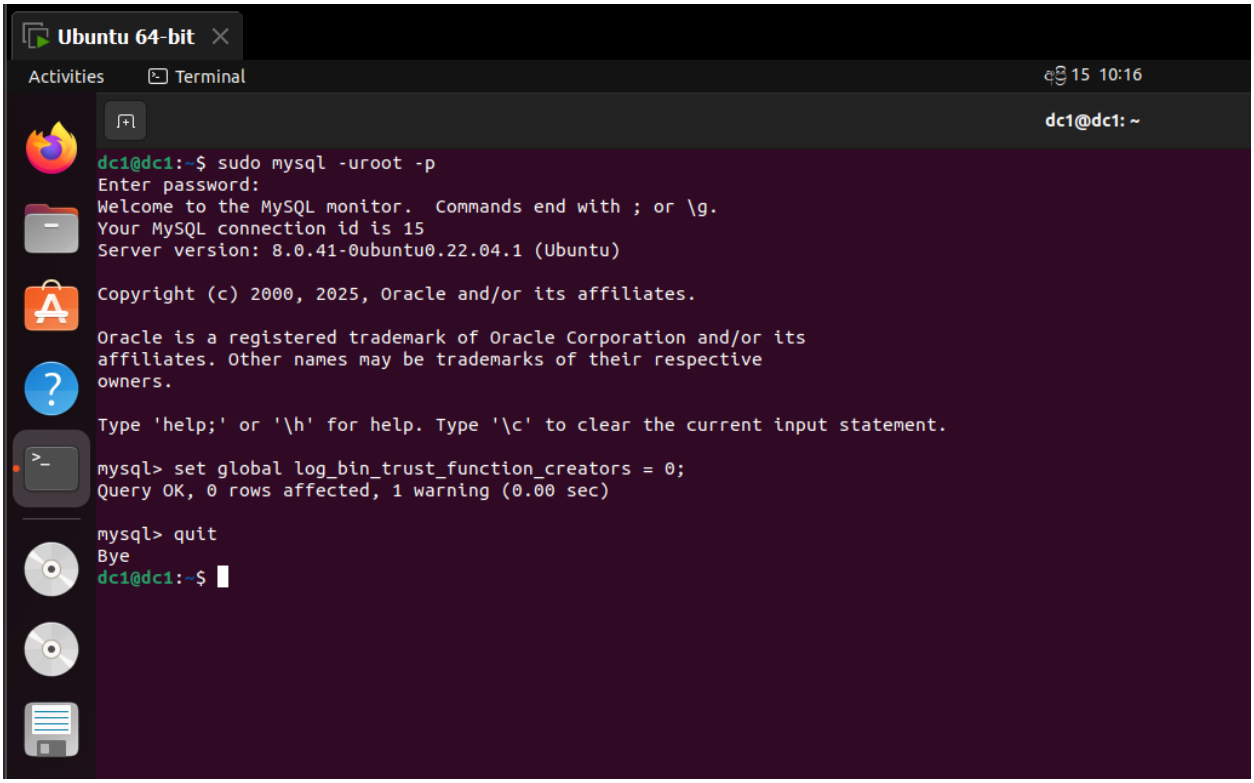


Figure 3.0.13-Disable the log_bin_trust_function_creators option after importing the database schema

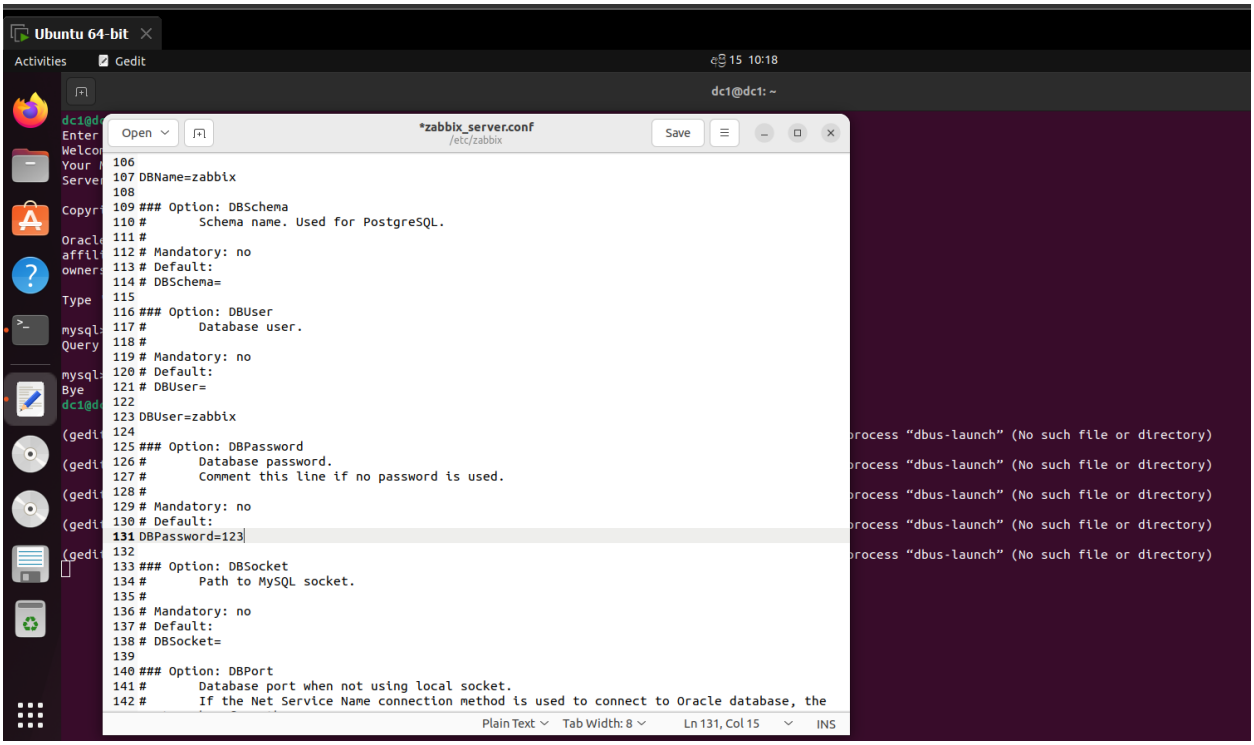


Figure 3.0.14-Edit file zabbix_server.conf

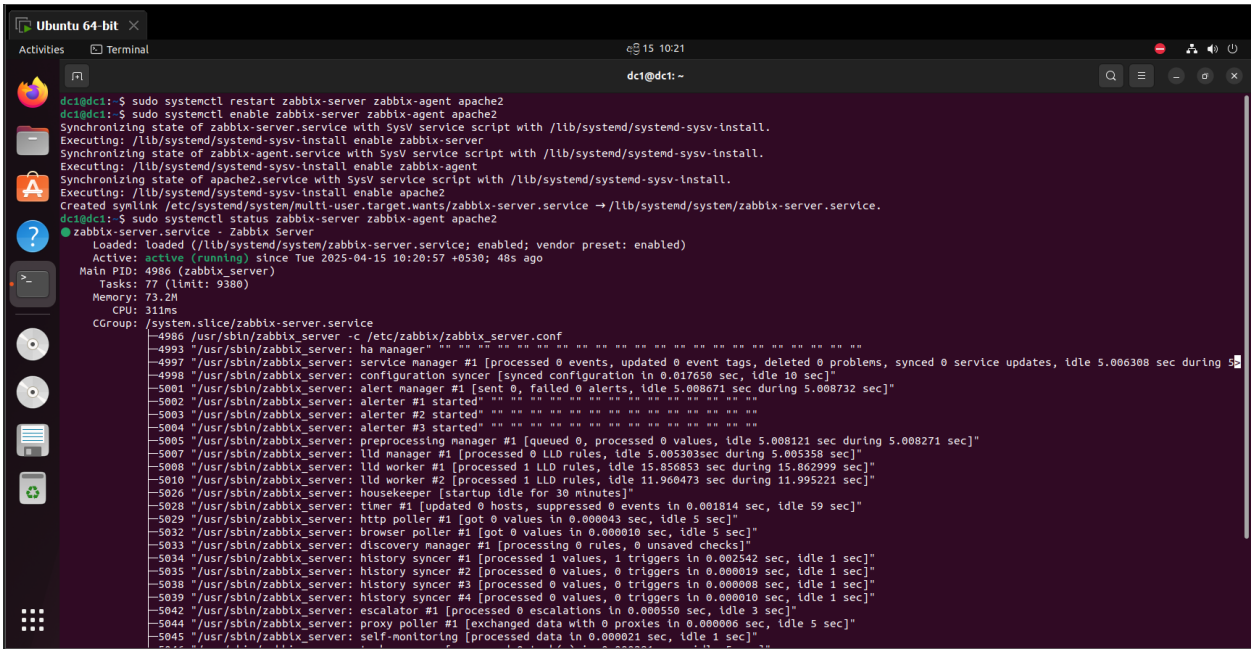


Figure 3.0.15-Start Zabbix server and agent processes

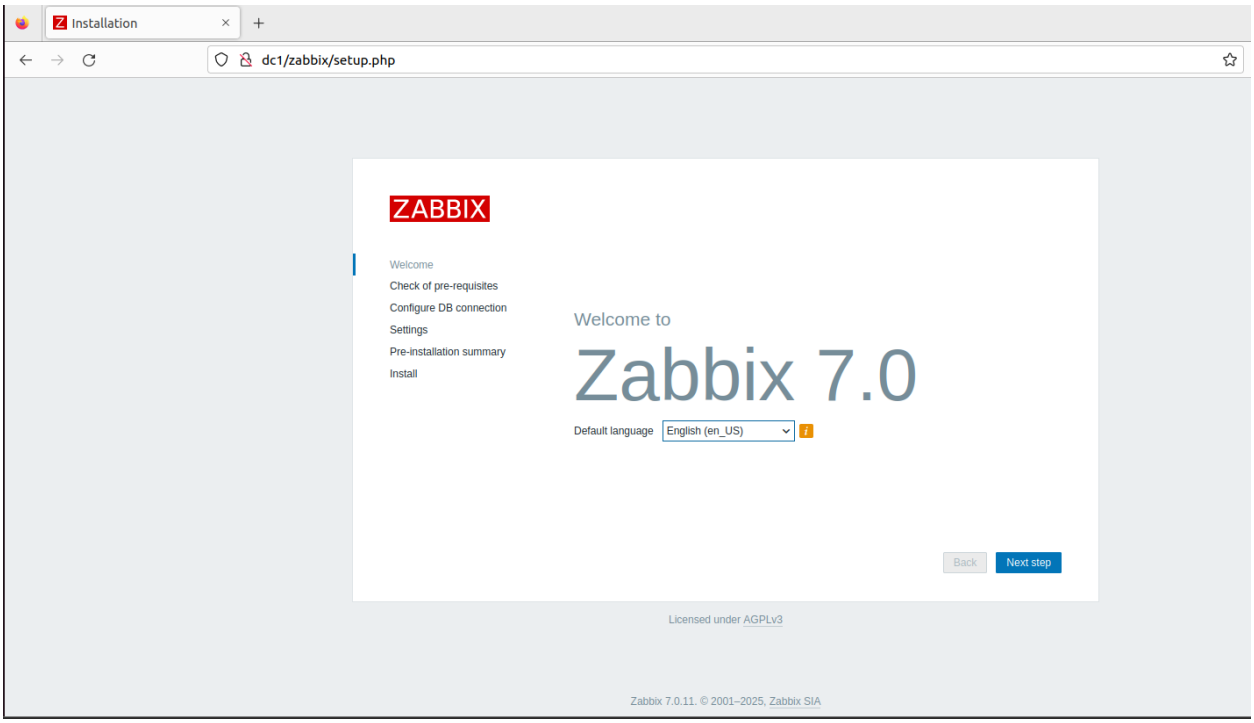


Figure 3.0.16-Search https://dc1/zabbix

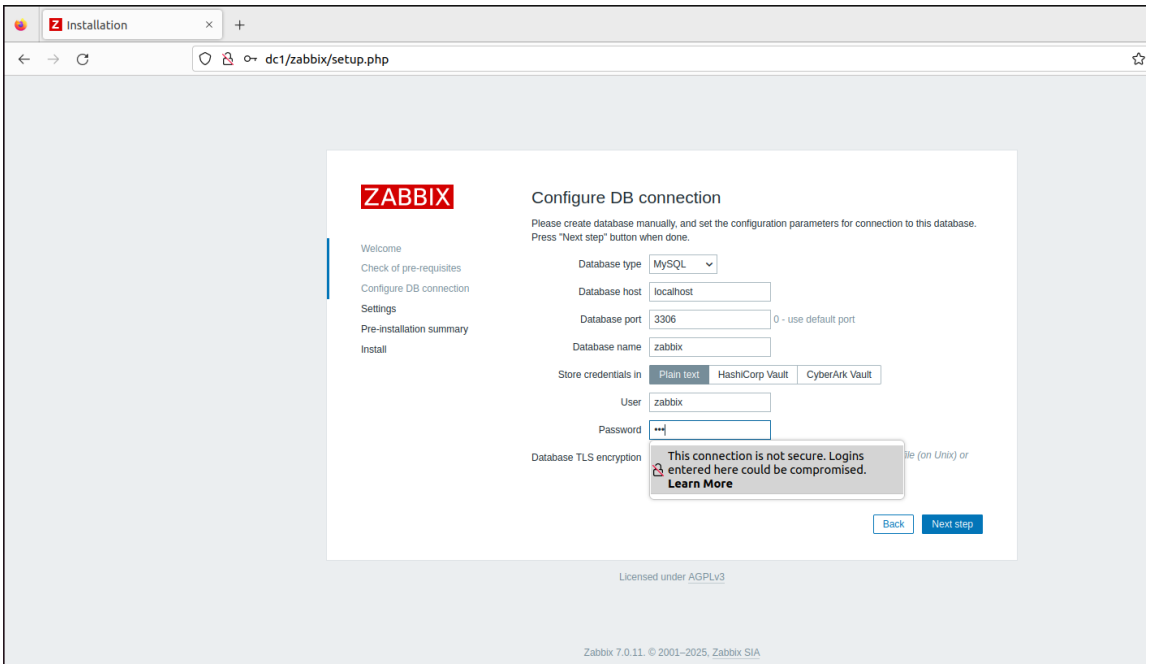


Figure 3.0.17-Configure Zabbix username, password and port

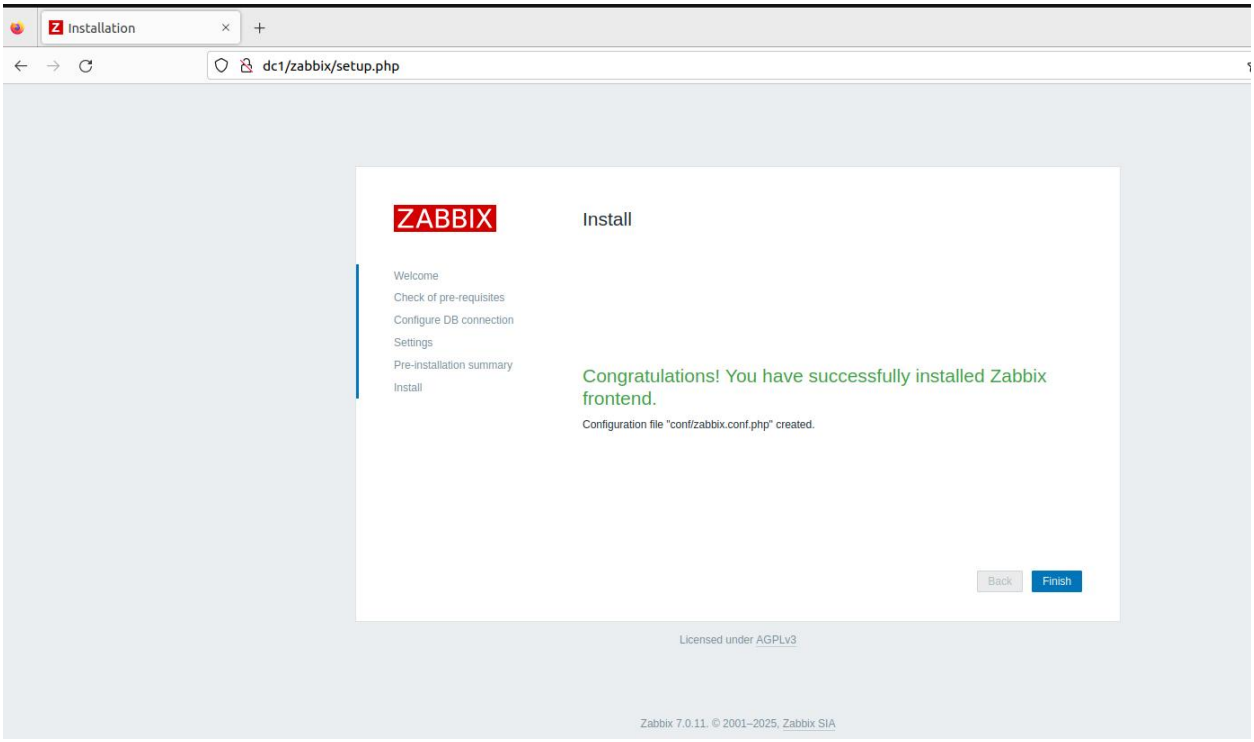


Figure 3.0.18-Successfully installed Zabbix server

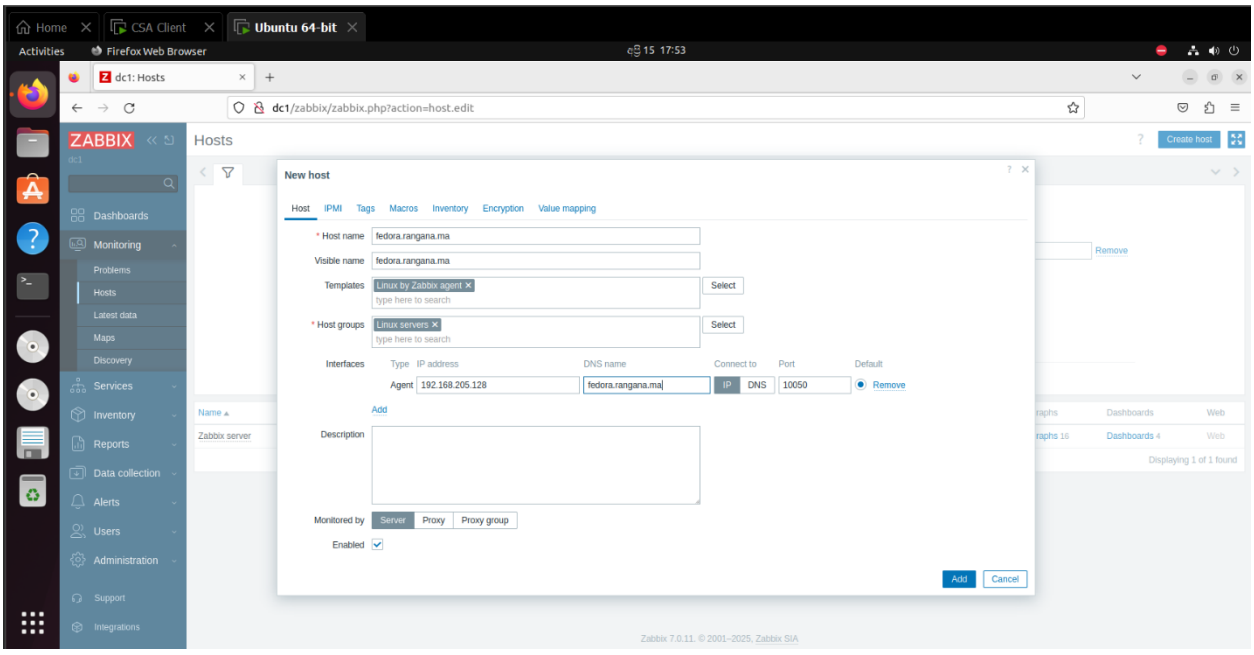


Figure 3.0.19-Create a new host using the Zabbix server

6. Install the Zabbix agent on the clients and connect to the Zabbix server

6.1 Connect the Windows client to Zabbix

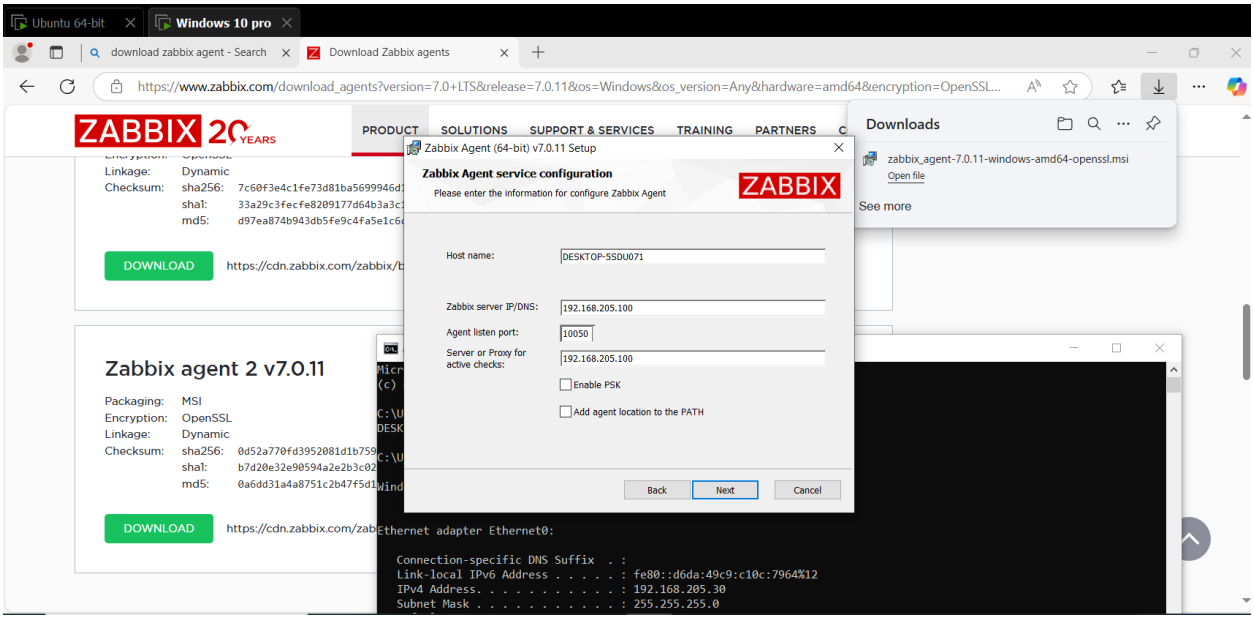


Figure 4.0.1-Add Zabbix agent

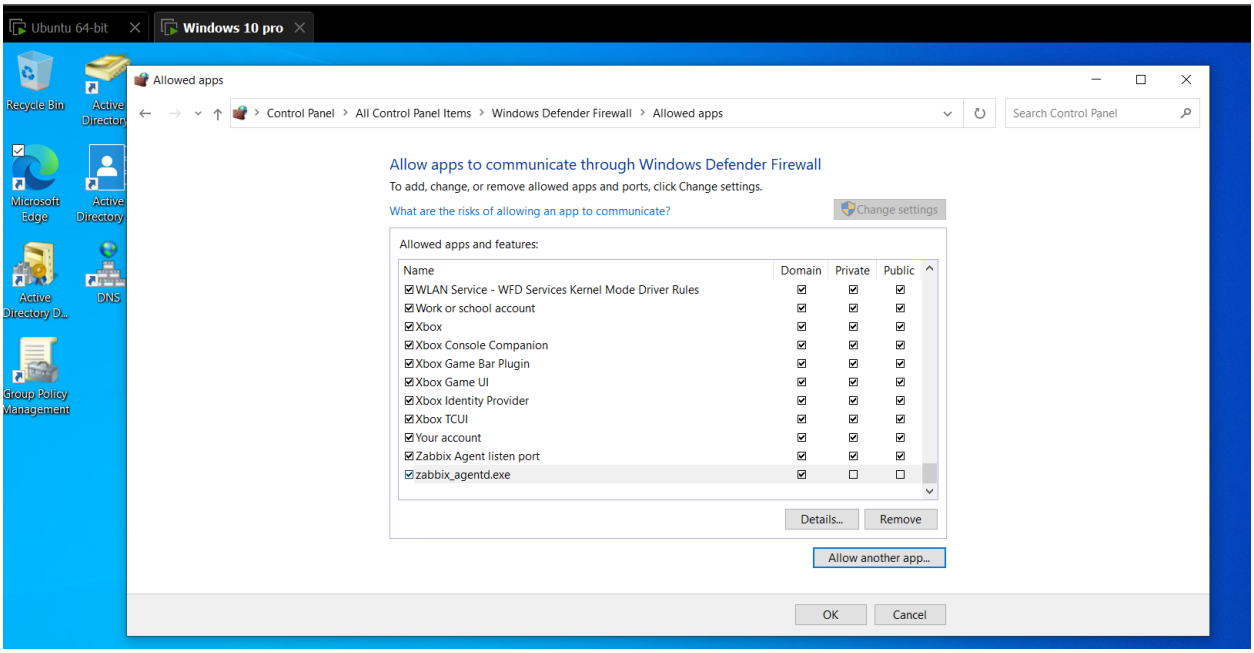
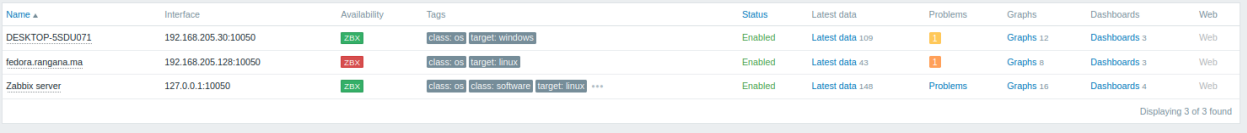


Figure 4.0.2-Give the firewall access to Zabbix agent

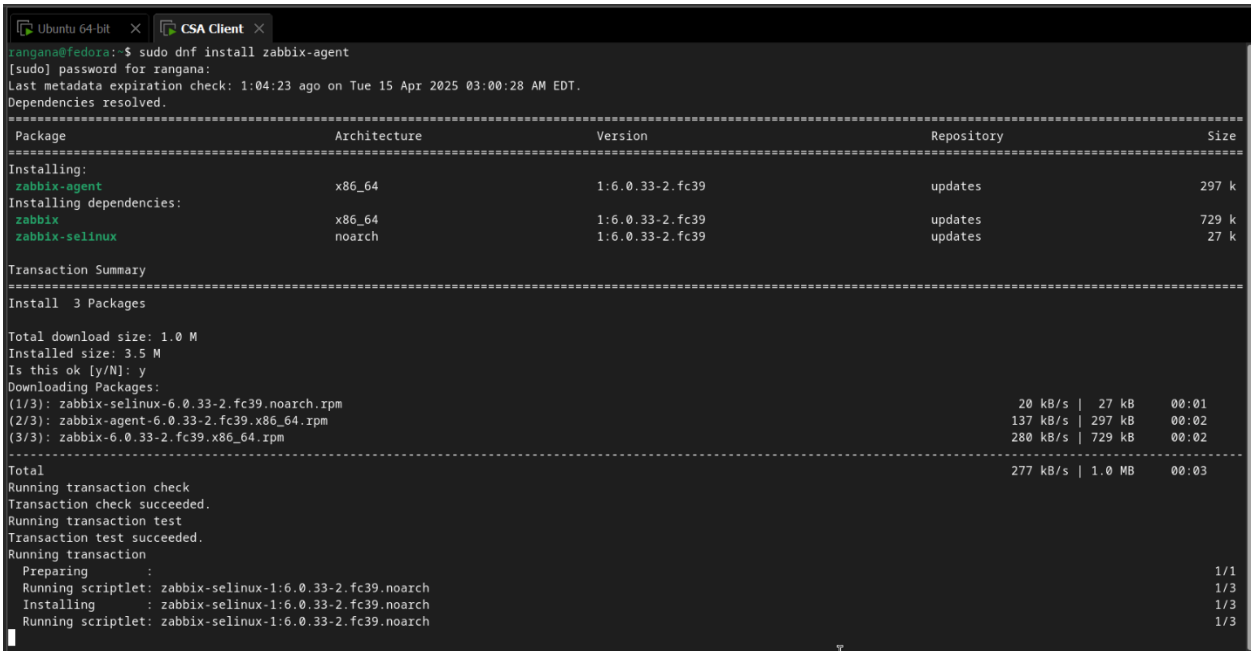


Name ▲	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
DESKTOP-5SDU071	192.168.205.30:10050	ZBX	class: os target: windows	Enabled	Latest data 109	1	Graphs 12	Dashboards 3	Web
fedora.rangana.ma	192.168.205.128:10050	ZBX	class: os target: linux	Enabled	Latest data 43	1	Graphs 8	Dashboards 3	Web
Zabbix server	127.0.0.1:10050	ZBX	class: os class: software target: linux •••	Enabled	Latest data 148	Problems	Graphs 16	Dashboards 4	Web

Displaying 3 of 3 found

Figure 4.0.3-Added Windows client to the Zabbix server

6.2 Connect the Fedora client to Zabbix



```
Ubuntu 64-bit   CSA Client
rangana@fedora:~$ sudo dnf install zabbix-agent
[sudo] password for rangana:
Last metadata expiration check: 1:04:23 ago on Tue 15 Apr 2025 03:00:28 AM EDT.
Dependencies resolved.
================================================================================
 Package                Architecture          Version                Repository          Size
================================================================================
Installing:
 zabbix-agent           x86_64                1:6.0.33-2.fc39        updates            297 k
Installing dependencies:
 zabbix                 x86_64                1:6.0.33-2.fc39        updates            729 k
 zabbix-selinux         noarch                1:6.0.33-2.fc39        updates             27 k

Transaction Summary
================================================================================
Install  3 Packages

Total download size: 1.0 M
Installed size: 3.5 M
Is this ok [y/N]: y
Downloading Packages:
(1/3): zabbix-selinux-6.0.33-2.fc39.noarch.rpm      20 kB/s |  27 kB     00:01
(2/3): zabbix-agent-6.0.33-2.fc39.x86_64.rpm       137 kB/s | 297 kB     00:02
(3/3): zabbix-6.0.33-2.fc39.x86_64.rpm             280 kB/s | 729 kB     00:02
--------------------------------------------------------------------------------
Total                                              277 kB/s | 1.0 MB     00:03
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: zabbix-selinux-1:6.0.33-2.fc39.noarch                  1/3
  Installing       : zabbix-selinux-1:6.0.33-2.fc39.noarch                  1/3
  Running scriptlet: zabbix-selinux-1:6.0.33-2.fc39.noarch                  1/3
```

Figure 4.0.4-Install Zabbix agent

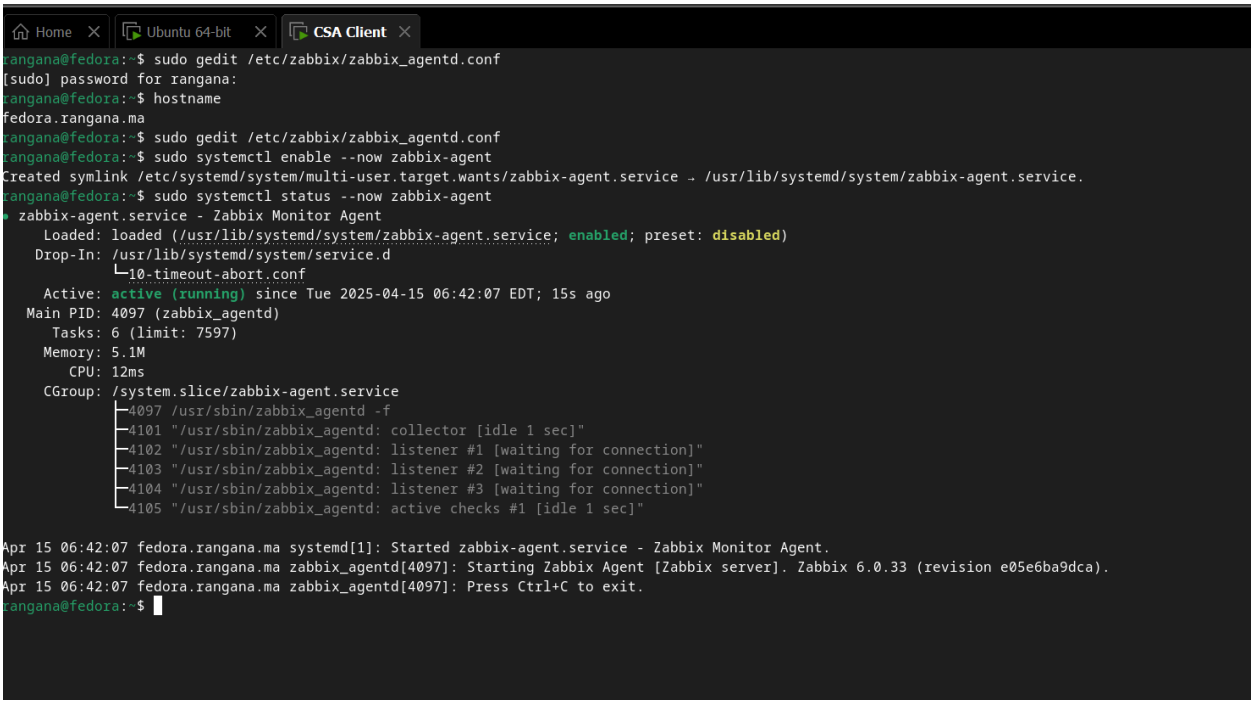


Figure 4.0.5-start and enable the agent

7. Errors Occurred During the Configuration Process

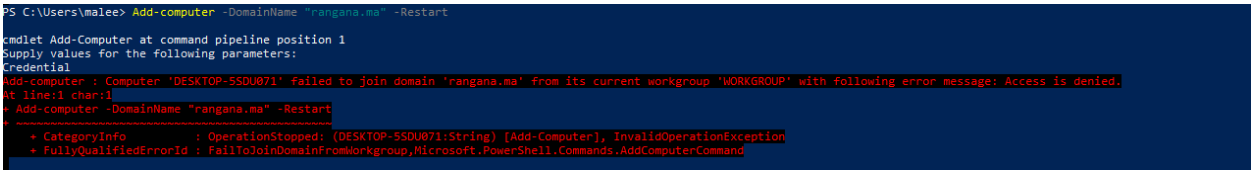


Figure 5.0.1-Issue with PowerShell syntax or session permission in Windows

This issue was caused by insufficient permissions in the PowerShell session or incorrect credentials. To resolve it, I switched to using the **Windows GUI method**:

1. Opened **System Properties** > **Computer Name** > **Change**.
2. Selected **Domain**, entered rangana.ma, and provided valid **domain administrator credentials**.
3. Successfully joined the domain and **restarted the system**.

This ensured the client was added to the Samba AD domain without further issues.

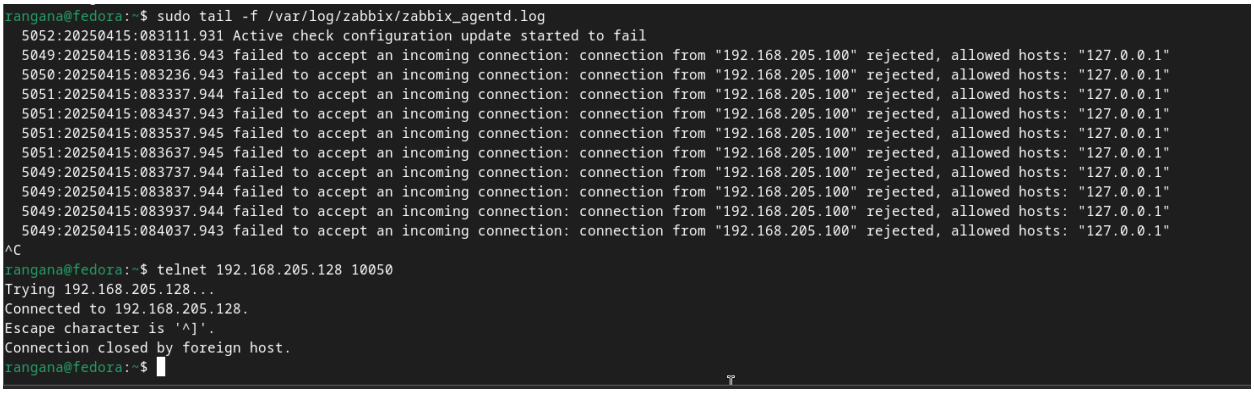


Figure 5.0.2-Try to fix the Zabbix agent

In the Zabbix agent configuration file (/etc/zabbix/zabbix_agentd.conf), the Server= and/or ServerActive= parameter was set to 127.0.0.1, which means the agent only accepted connections from the local machine.

Resolution Steps:

1. **Edit the Zabbix Agent Configuration:**

   sudo nano /etc/zabbix/zabbix_agentd.conf

2. **Update the Server and ServerActive parameters:**

   Server=127.0.0.1

   ServerActive=127.0.0.1

   To:

   Server=127.0.0.1,192.168.205.100

   ServerActive=192.168.205.100

This allows passive checks from both localhost and your Zabbix server, and active checks to the Zabbix server.

3. **Restart the Zabbix Agent:**

   sudo systemctl restart zabbix-agent

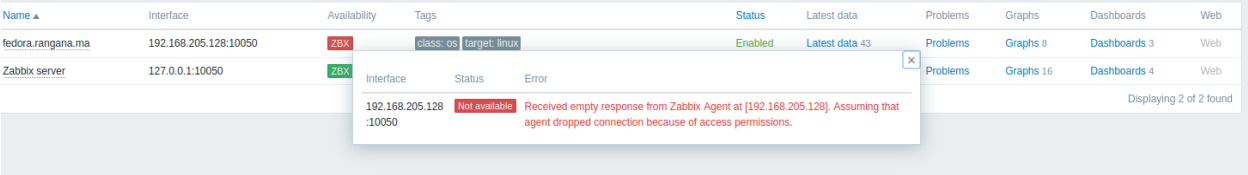


Figure 5.3-Zabbix Agent on Fedora machine is not responding to server messages

In the agent configuration file (/etc/zabbix/zabbix_agentd.conf), the Server= parameter was limited to 127.0.0.1, so it rejected connections from the actual Zabbix server's IP (192.168.205.100).

- **Open the agent config file:**

  sudo nano /etc/zabbix/zabbix_agentd.conf

- **Update the access control parameters:**

  Server=127.0.0.1

  ServerActive=127.0.0.1

  After:

  Server=127.0.0.1,192.168.205.100

  ServerActive=192.168.205.100

This allows the Zabbix server (at 192.168.205.100) to perform both passive and active checks.

- **Restart the agent to apply changes:**

  sudo systemctl restart zabbix-agent

- **Verify Firewall Rules:**

Make port 10050 is open to the Zabbix server:

sudo firewall-cmd --add-port=10050/tcp --permanent

sudo firewall-cmd --reload

- **Check Agent Logs for Confirmation:**

  sudo tail -f /var/log/zabbix/zabbix_agentd.log

8. References

1. Samba Team, *Samba Documentation*. [Online]. Available: https://www.samba.org/samba/docs/
2. Zabbix LLC, *Zabbix Documentation 6.0 LTS*. [Online]. Available: https://www.zabbix.com/documentation/6.0/manual
3. Red Hat, "How to configure Samba on Linux," *Red Hat Customer Portal*, 2023. [Online]. Available: https://access.redhat.com/solutions/40492
4. Zabbix LLC, "Installing Zabbix Agent on Linux," *Zabbix Documentation 6.0*, 2023. [Online]. Available: https://www.zabbix.com/documentation/6.0/en/manual/installation/install_from_package
5. Fedora Project, "Firewalld Documentation," *Fedora Docs*, 2023. [Online]. Available: https://docs.fedoraproject.org/en-US/quick-docs/firewalld/